DCCN Project

Course Title:

Data Communication and Computer Networks

Lab Report Title:

Creating Final Project

Submitted to:

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Submitted by:

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Registration No:

(22245, 22292 , 22068 , 22105)

Program:

BSCS- 5th Semester

RIP using DHCP : (Stage 1)

We are using 4 networks (LANS) with default gateways as mentioned below :

They have starting IP addresses as mentioned below :

LAN 1 :

Default Gateway : 192.168.1.1

LAN 2 :

Default Gateway : 215.147.22.1

LAN 3 :

Default Gateway : 192.168.0.1

LAN 4 :

Default Gateway : 172.16.1.1

LAN 1 :

Starting IPs : 192.168.1.151

LAN 2 :

Starting IPs : 215.147.22.15

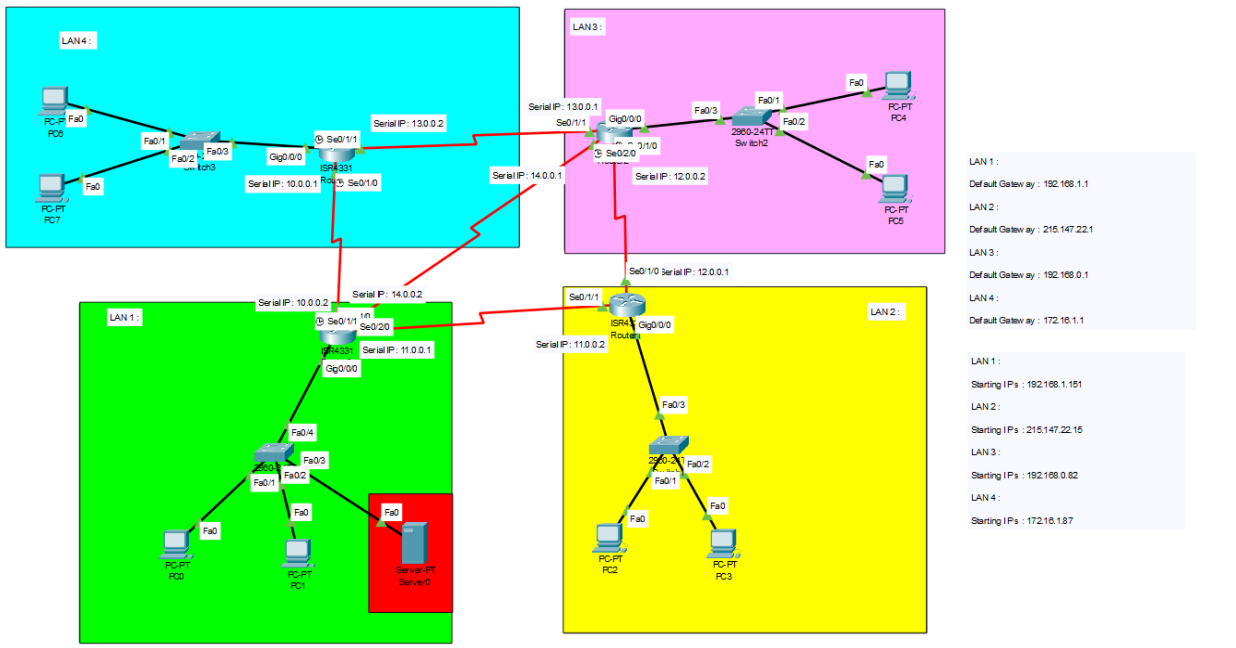
LAN 3 :

Starting IPs : 192.168.0.82

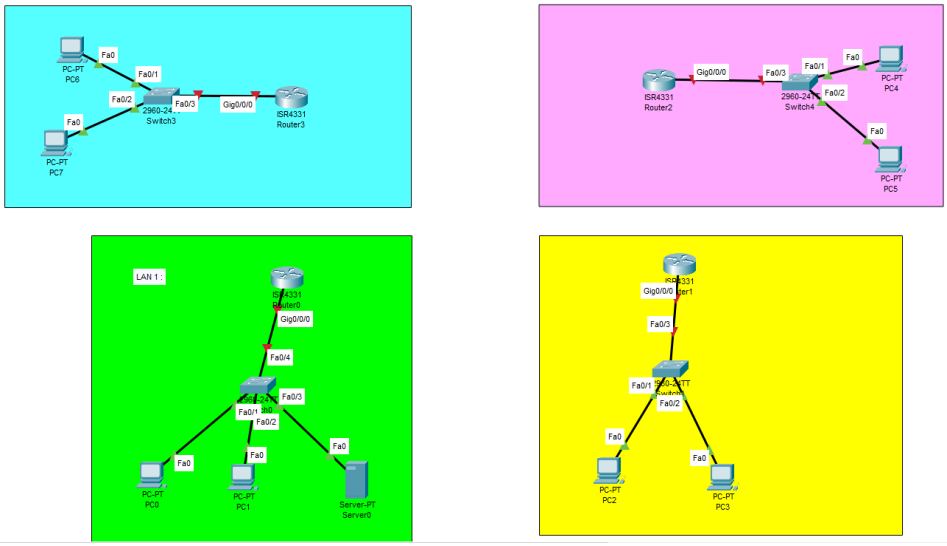
LAN 4 :

Starting IPs : 172.16.1.87

Final Topology of DHCP(RIP) :

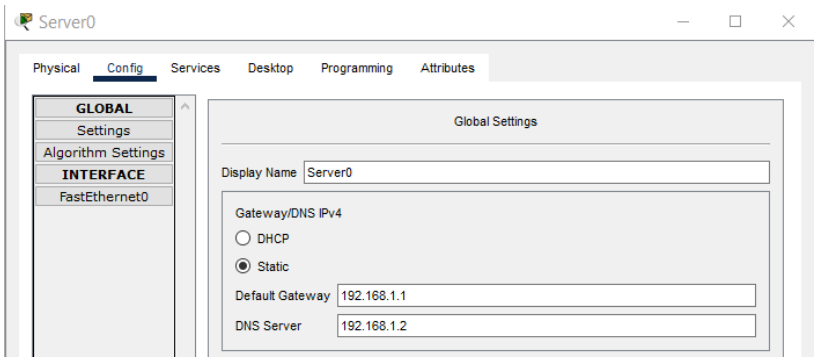


Step 1 : We have created basic topologies of 4 networks (LANS) as shown

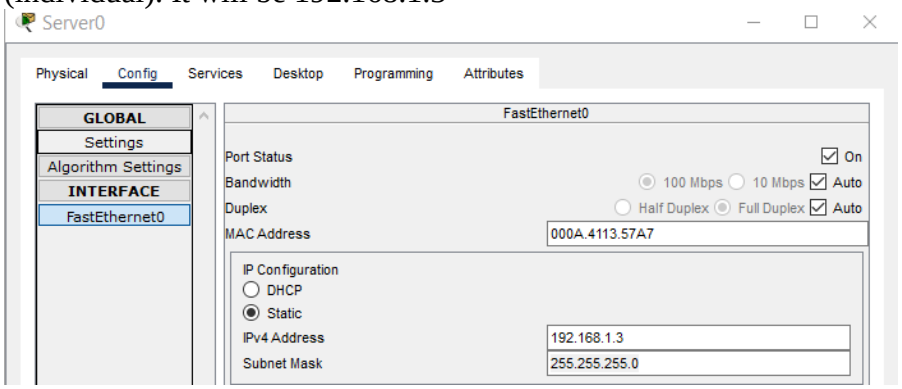


1. Go to server :

Click on server and go to Config write the default gateways as 192.168.1.1 and write the DNS server as 192.168.1.2



2. Go to FastEthernet0 and gives IPv4 Address as 192.168.1.3 it is the Server's own IP address (individual). It will be 192.168.1.3



3. Go to Services in DHCP Server :

For Network (LAN) 1 Its name will be serverPool :

Write the default gateway similar to previous 192.168.1.1 & DNS server IP address similar to previous 192.168.1.2

Write starting IP address :

We are assigned with IP addresses starting from 192.168.1.151, It belongs to Class C so, we will write its subnet mask 255.255.255.0, Save and add it.

For Network (LAN) 2 Its name will be serverPool_A :

Write the default gateway of LAN 2 215.147.22.1 & DNS server IP address same 192.168.1.2

Write starting IP address :

We are assigned with IP addresses starting from 215.147.22.15, It belongs to Class C so, we will write its subnet mask 255.255.255.0, Save and add it.

For Network (LAN) 3 Its name will be serverPool_B :

Write the default gateway of LAN 3 192.168.0.1 & DNS server IP address remains same 192.168.1.2

Write starting IP address :

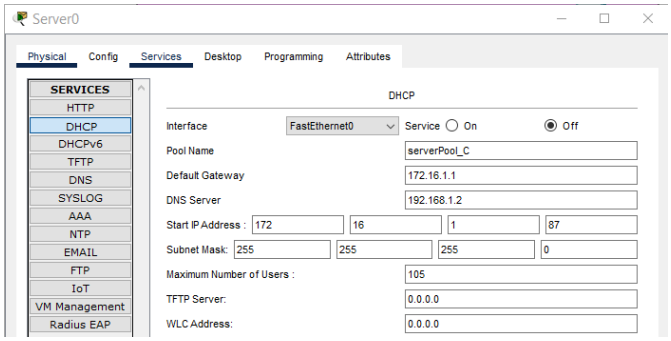
We are assigned with IP addresses starting from 192.168.0.82, It belongs to Class C so, we will write its subnet mask 255.255.255.0, Save and add it.

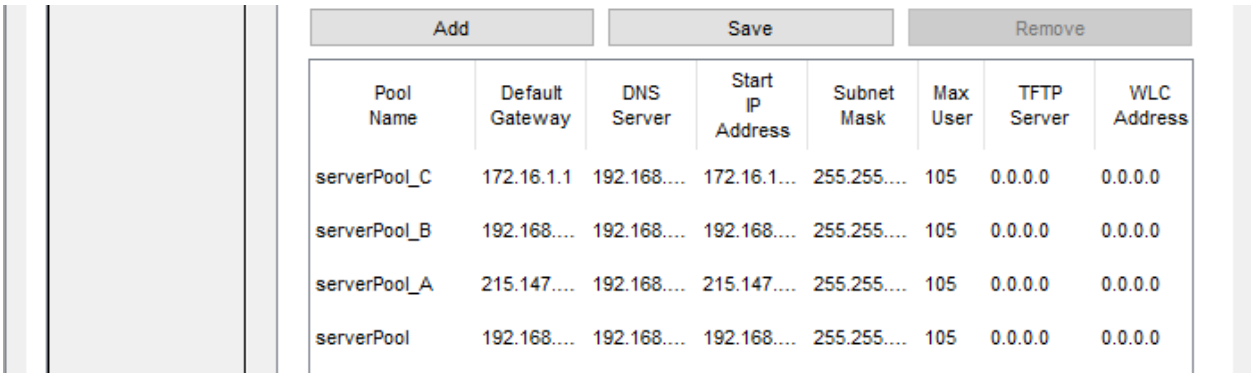
For Network (LAN) 4 Its name will be serverPool_C :

Write the default gateway of LAN 4 172.16.1.1 & DNS server IP address remains same 192.168.1.2

Write starting IP address :

We are assigned with IP addresses starting from 172.16.1.87, It belongs to Class B so, we will write its subnet mask 255.255.0.0, Save and add it.





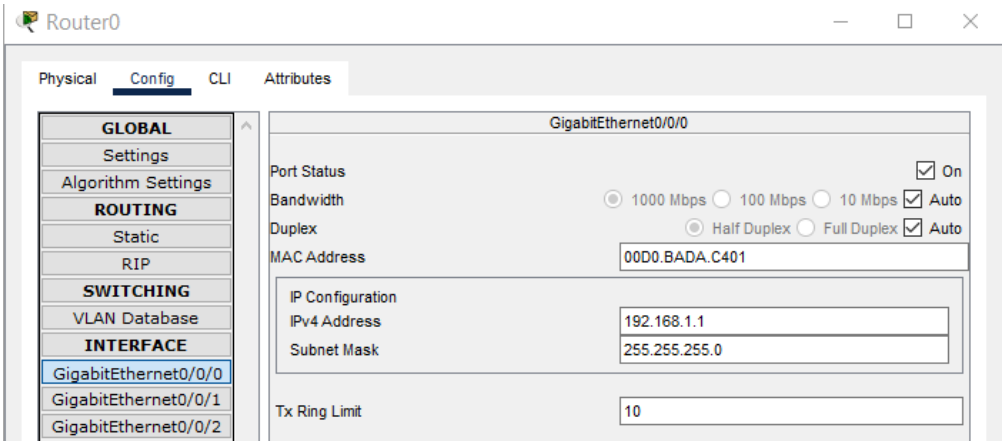
Pool Name	Default Gateway	DNS Server	Start IP Address	Subnet Mask	Max User	TFTP Server	WLC Address
serverPool_C	172.16.1.1	192.168....	172.16.1...	255.255....	105	0.0.0.0	0.0.0.0
serverPool_B	192.168....	192.168....	192.168....	255.255....	105	0.0.0.0	0.0.0.0
serverPool_A	215.147....	192.168....	215.147....	255.255....	105	0.0.0.0	0.0.0.0
serverPool	192.168....	192.168....	192.168....	255.255....	105	0.0.0.0	0.0.0.0

Step 2 :

Go to routers in Config (gig) and assign the IP addresses (Default Gateways) of their connected networks.

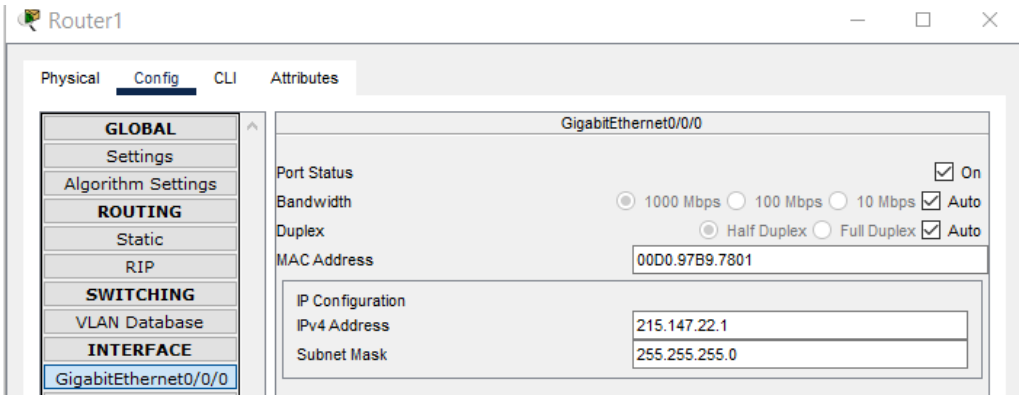
In Router 0 :

Give the Default gateway : 192.168.1.1 and turn it on.



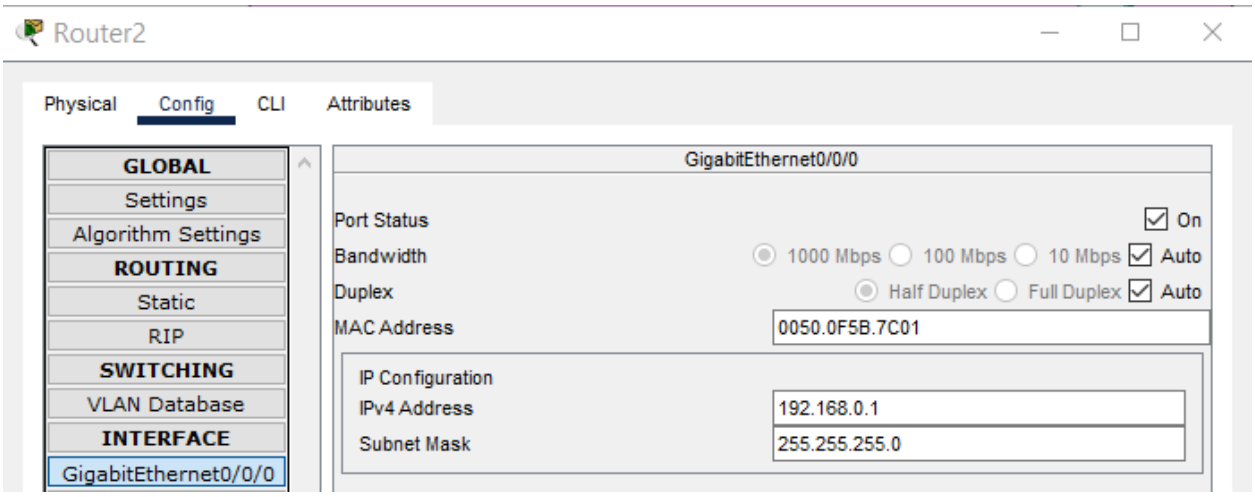
In Router 1 :

Give the Default gateway : 215.147.22.1 and turn it on.



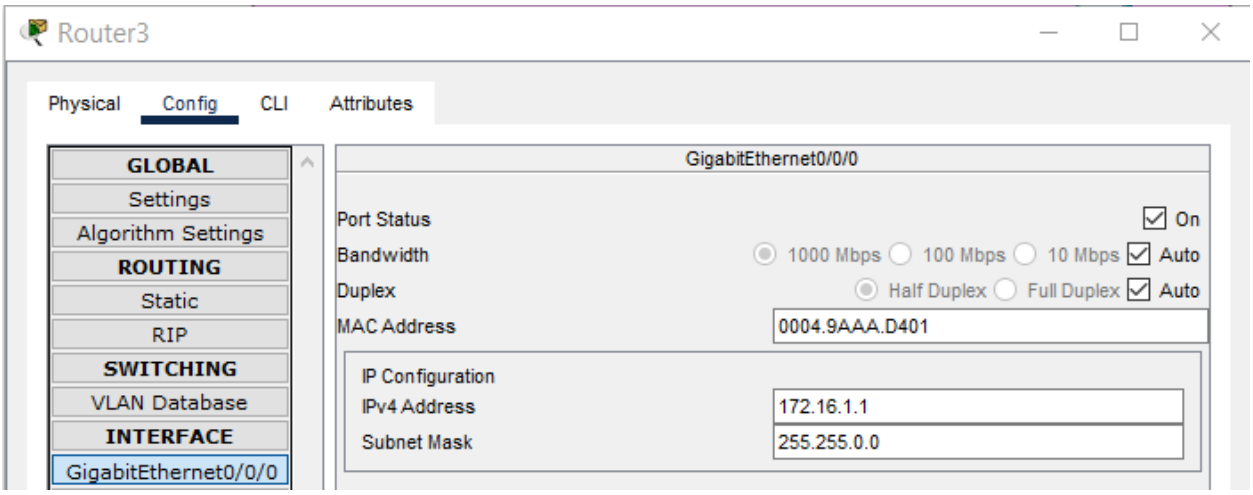
In Router 2 :

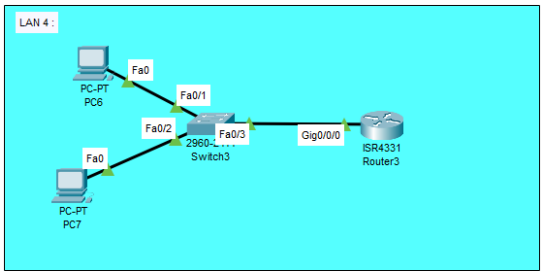
Give the Default gateway : 192.168.0.1 and turn it on.

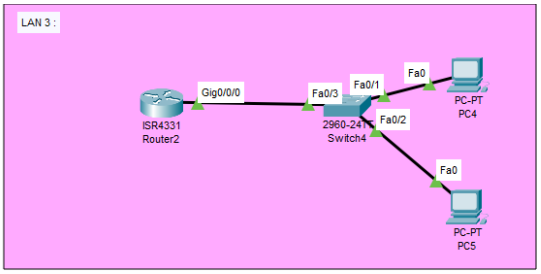


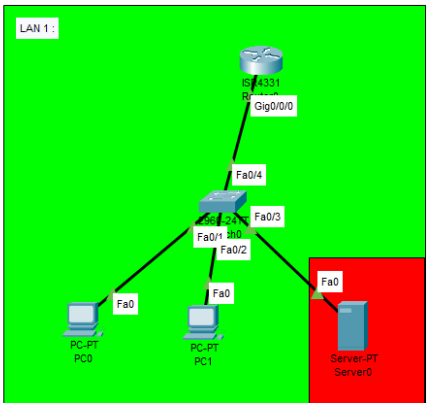
In Router 3 :

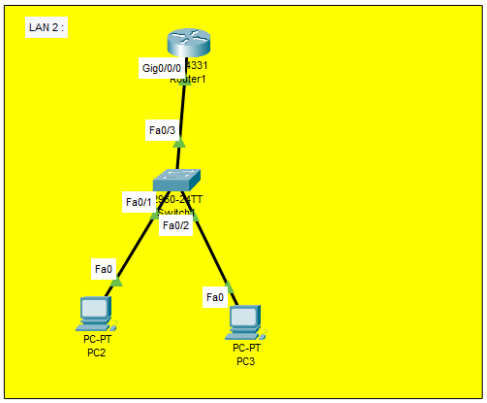
Give the Default gateway : 172.16.1.1 and turn it on.











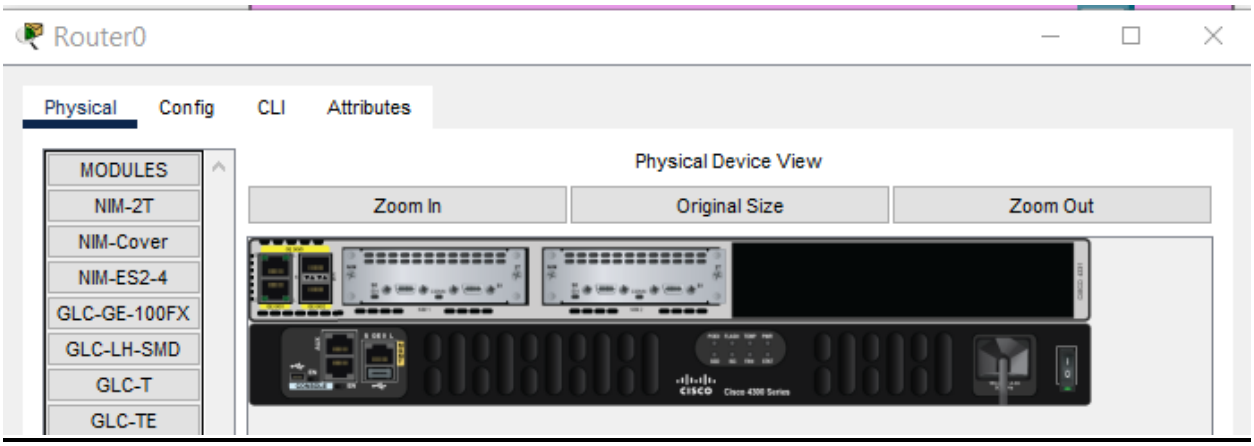
As it can be seen in the picture that we have 4 networks (LANS) in each network there is switch and with each switch there are 2 PCs connected. In 1 Network (LAN) there is DHCP Server which will automatically assign the IP addresses to all PCs once the Routers are connected and configured. The green arrow shows that the PCs with Switches and Switches with Routers are configured correctly.

Step 3 :

Add Module in each Router.

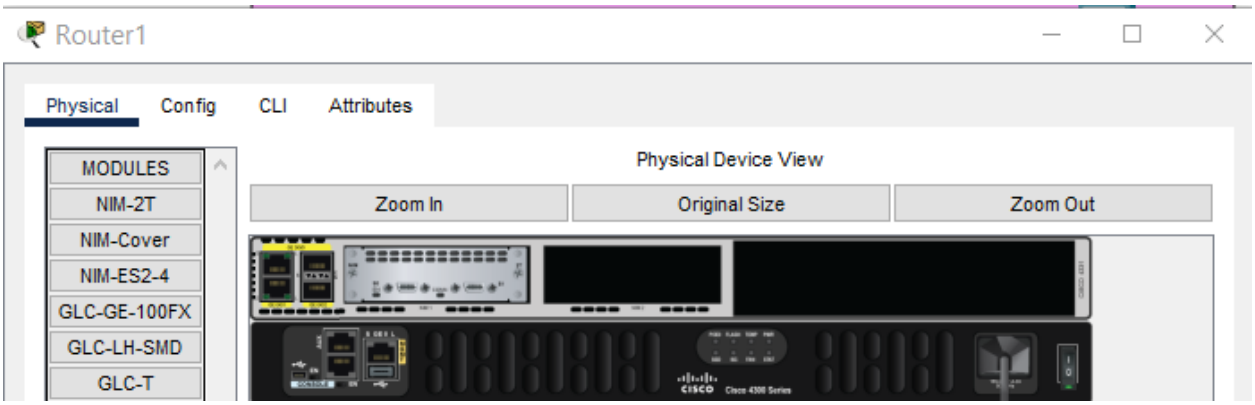
In Router 0 :

Add 2 Modules



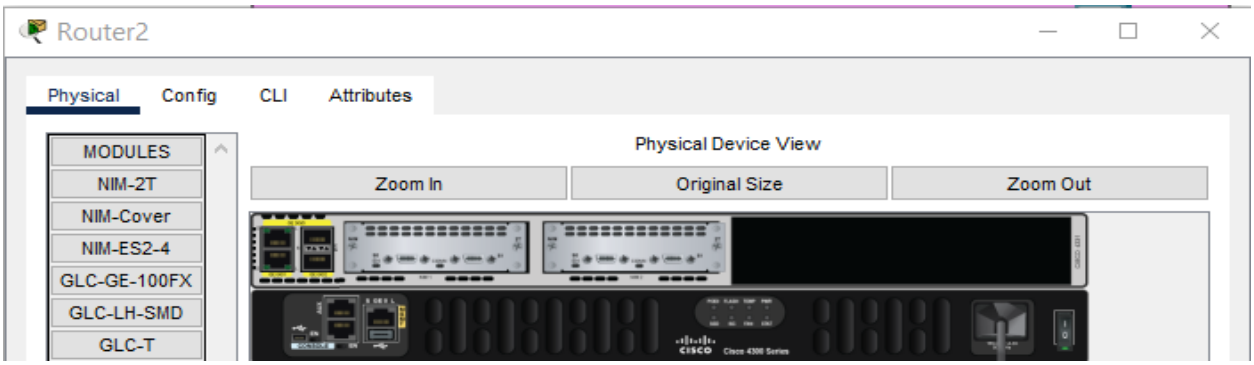
In Router 1 :

Add 1 Module



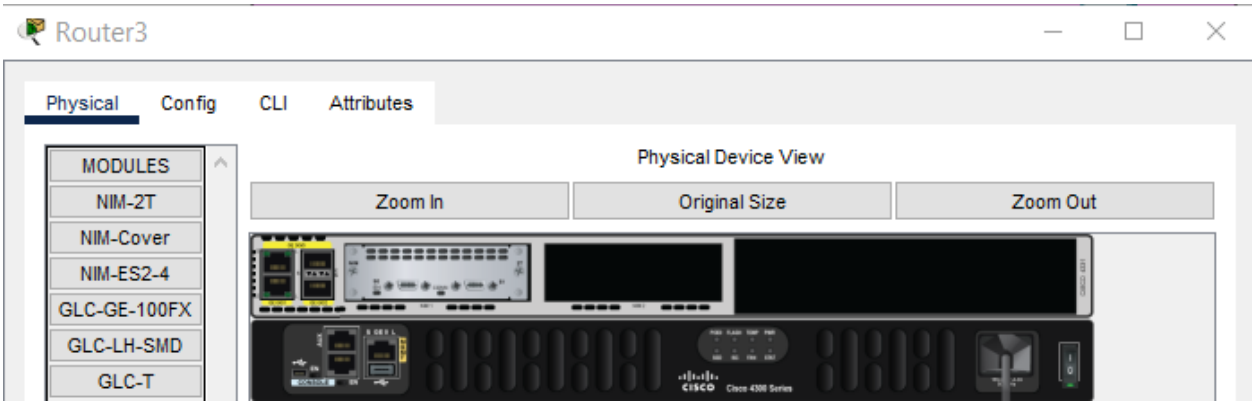
In Router 2 :

Add 2 Modules



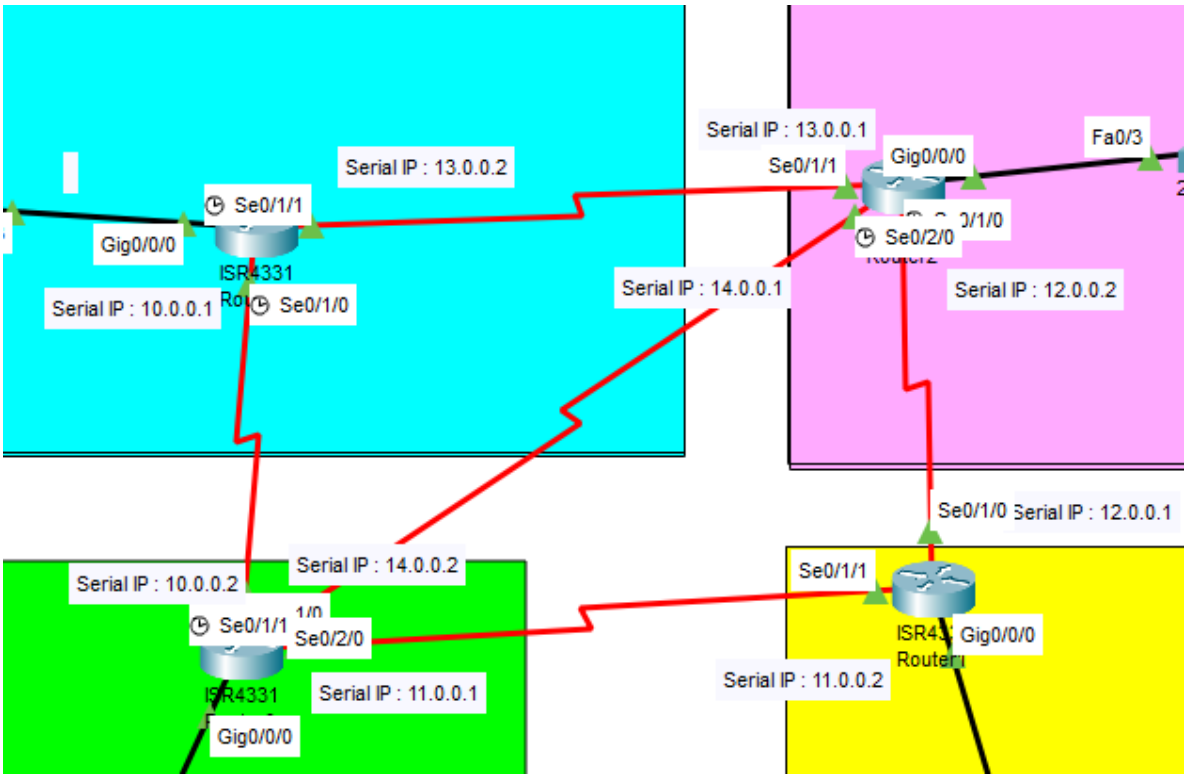
In Router 3 :

Add 1 Module



Assigning IP addresses to Serial Ports :

Clear Picture of Serial Port numbers along their IP addresses are written with them.



In Router 0 :

Gives Serial Port 0/1/0 IP address: 10.0.0.2

Gives Serial Port 0/1/1 IP address: 11.0.0.1

Gives Serial Port 0/2/0 IP address: 14.0.0.2

In Router 1 :

Gives Serial Port 0/1/1 IP address: 11.0.0.2

Gives Serial Port 0/1/0 IP address: 12.0.0.1

In Router 2 :

Gives Serial Port 0/1/1 IP address: 13.0.0.1

Gives Serial Port 0/1/0 IP address: 12.0.0.2

Gives Serial Port 0/2/0 IP address: 14.0.0.1

In Router 3 :

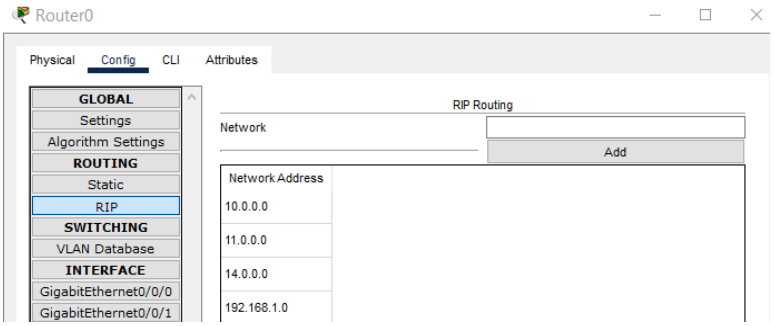
Gives Serial Port 0/1/0 IP address: 10.0.0.1

Gives Serial Port 0/1/1 IP address: 13.0.0.2

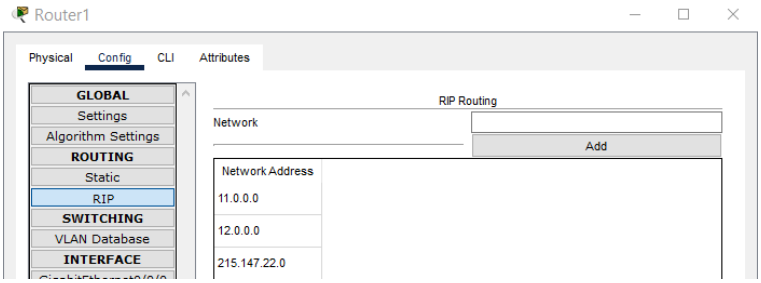
Step 4 :

On all Routers go to Config (RIP) and add all the networks the Router is connected with. Write the Network of all wires it (Router) is connected to.

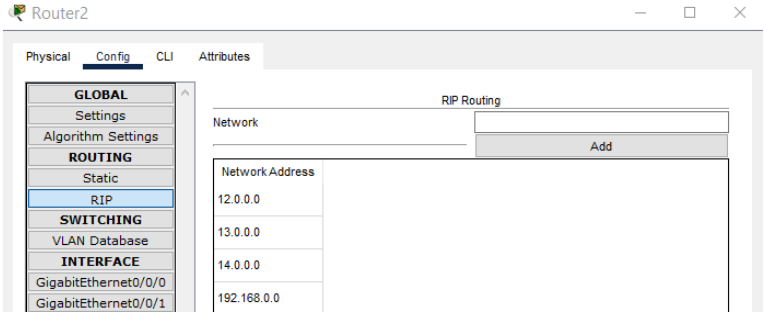
In Router 0: (RIP Networks)



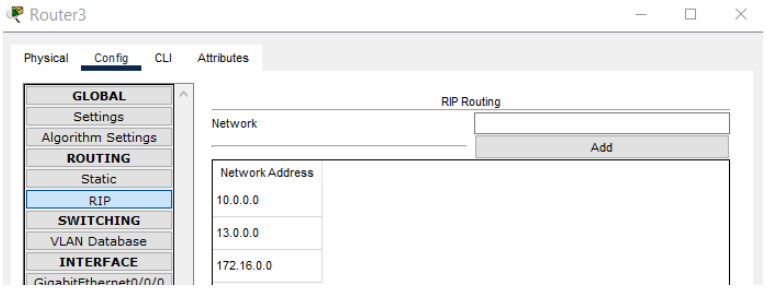
In Router 1 : (RIP Networks)



In Router 2 : (RIP Networks)



In Router 3 : (RIP Networks)



Step 5 : On all Routers :

Add 2 times exit.
en
conf t
int gig0/0/0
ip helper-address 192.168.1.3
Enter 2 times exit again.

On Router 0 add CLI commands as shown in the picture :

```
Router>en
Router#
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/2/0, changed state to up
c
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/1/0, changed state to up
Router#
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#int gig0/0/0
Router(config-if)#ip helper-address 192.168.1.3
Router(config-if)#exit
Router(config)#exit
```

On Router 1 add CLI commands as shown in the picture :

```
Router>en
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#int gig0/0/0
Router(config-if)#ip helper-address 192.168.1.3
Router(config-if)#exit
Router(config)#exit
Router#
```

On Router 2 add CLI commands as shown in the picture :

```
Router>en
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#int gig0/0/0
Router(config-if)#ip helper-address 192.168.1.3
Router(config-if)#exit
Router(config)#exit
```

On Router 3 add CLI commands as shown in the picture :

```
Router>en
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#int gig0/0/0
Router(config-if)#ip helper-address 192.168.1
                                    ^
% Invalid input detected at '^' marker.

Router(config-if)#ip helper-address 192.168.1.3
Router(config-if)#exit
Router(config)#exit
```

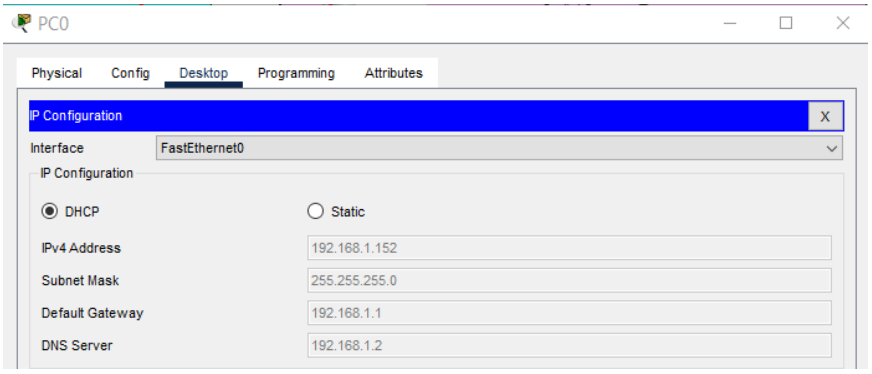
Step 6 :

On each PC go to Desktop then go to IP Configuration and click on DHCP.

The DHCP Server will assign all the PCs their IP Addresses automatically according to their start IP Addresses that we assigned it.

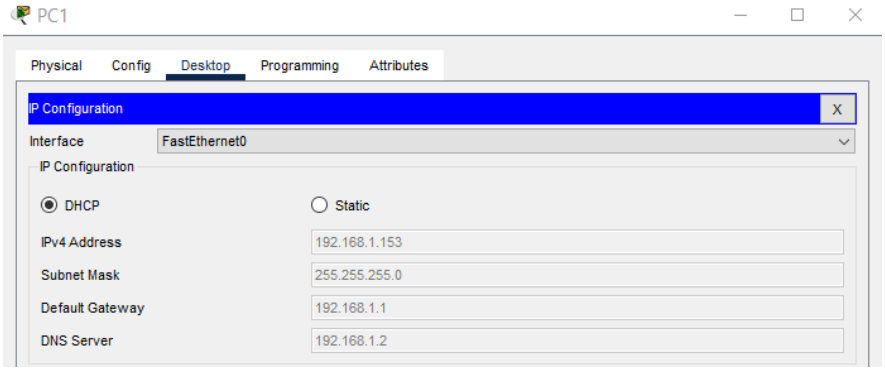
In PC 0 :

IPv4 Address = 192.168.1.152 (that is assigned by the DHCP Server automatically).



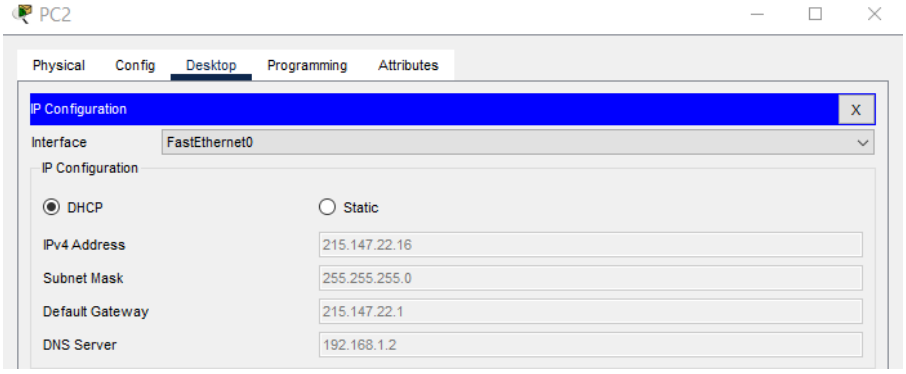
In PC 1 :

IPv4 Address = 192.168.1.153 (that is assigned by the DHCP Server automatically).



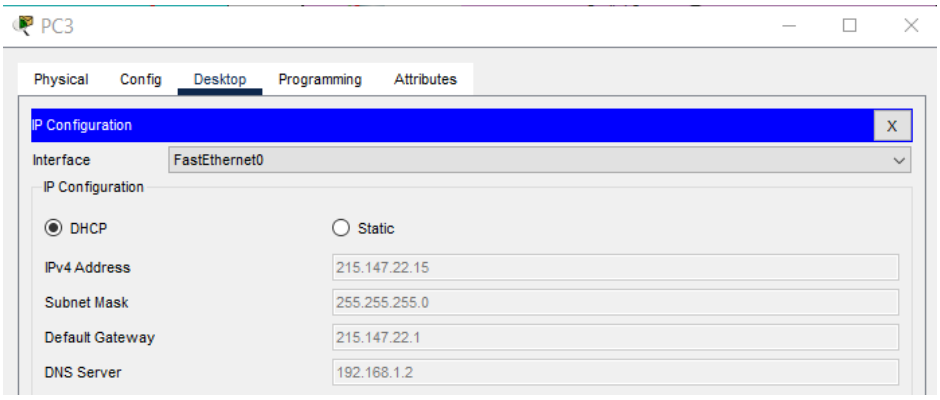
In PC 2 :

IPv4 Address = 215.147.22.16 (that is assigned by the DHCP Server automatically).



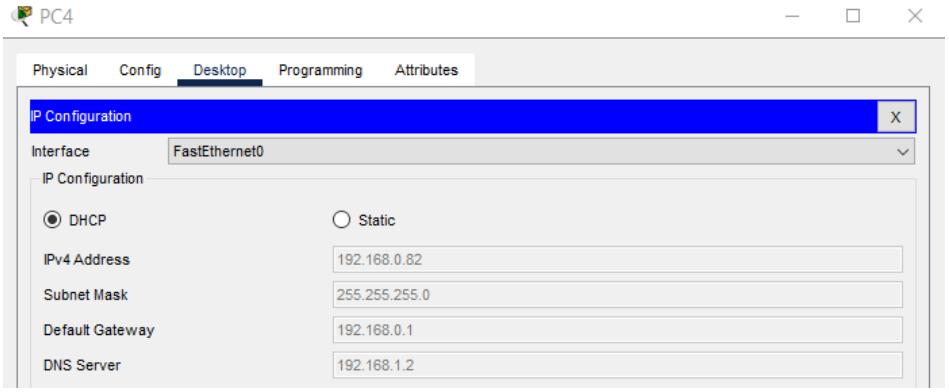
In PC 3 :

IPv4 Address = 215.147.22.15 (that is assigned by the DHCP Server automatically).



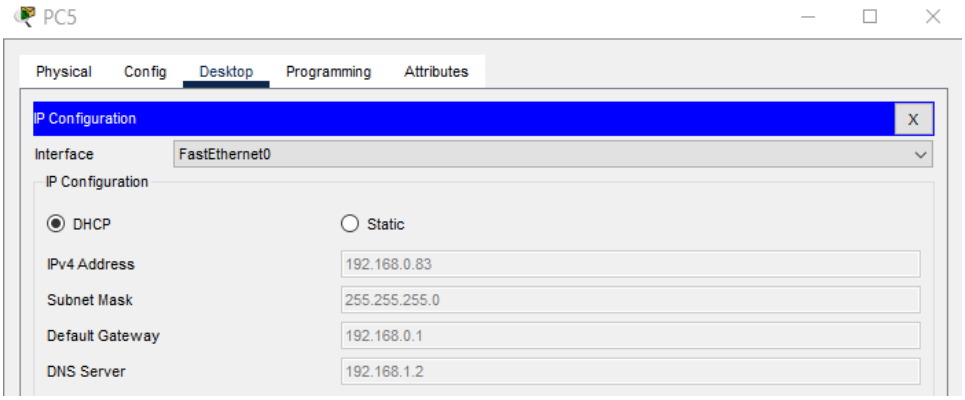
In PC 4 :

IPv4 Address = 192.168.0.82 (that is assigned by the DHCP Server automatically).



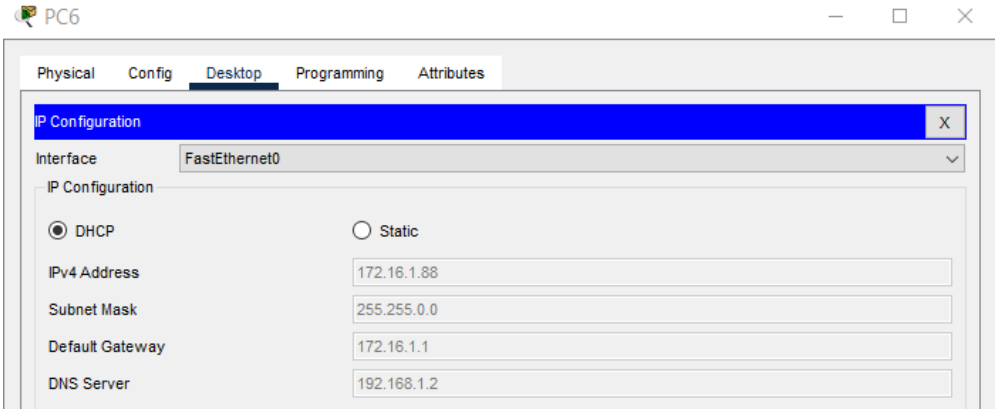
In PC 5 :

IPv4 Address = 192.168.0.83 (that is assigned by the DHCP Server automatically).



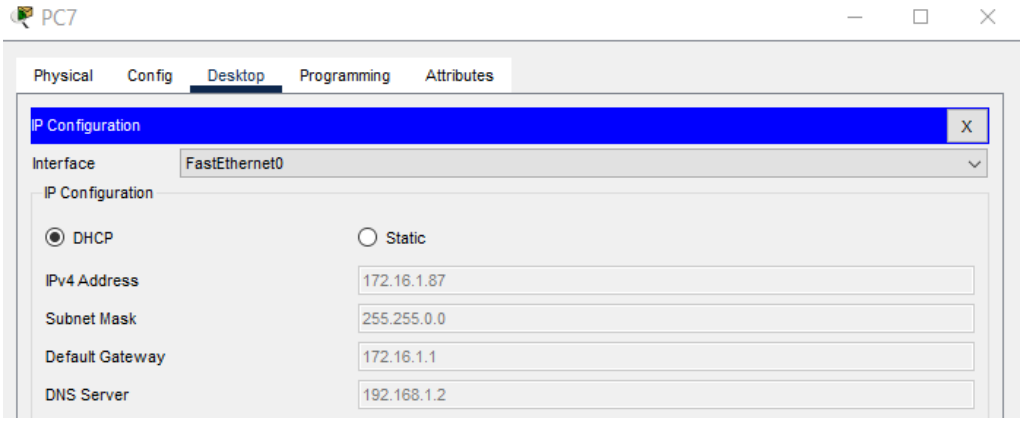
In PC 6 :

IPv4 Address = 172.16.1.88 (that is assigned by the DHCP Server automatically).



In PC 7 :

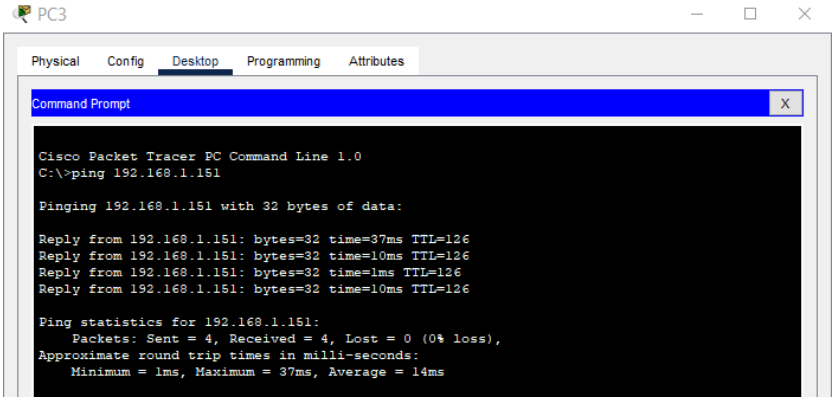
IPv4 Address = 172.16.1.87 (that is assigned by the DHCP Server automatically).



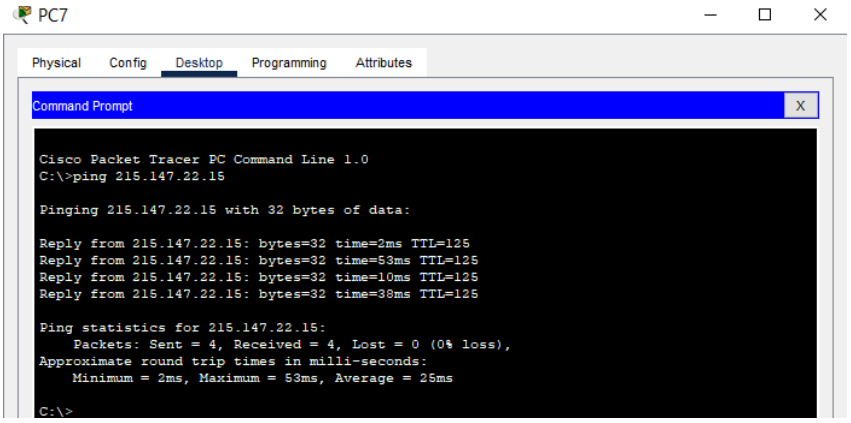
Step 7 :

Now we will Ping from PC (in 1 network) to another PC (in other network):

Ping from PC 3 (in LAN 2) to PC 0 (in LAN 1):

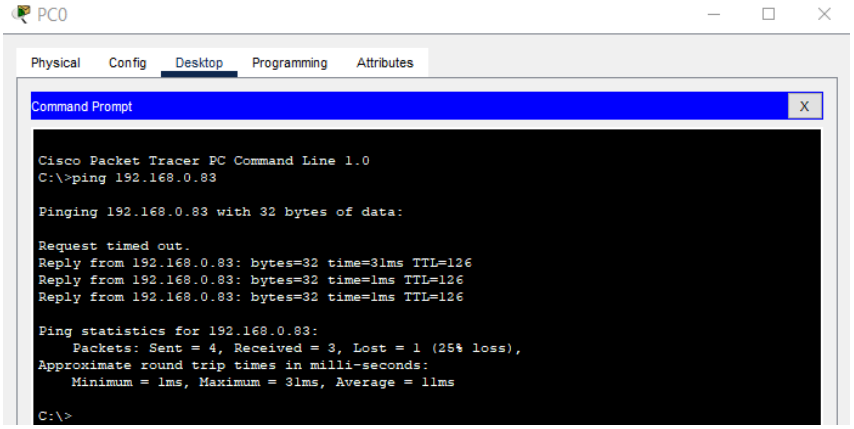


Ping from PC 3 (in LAN 2) to PC 7 (in LAN 4) :



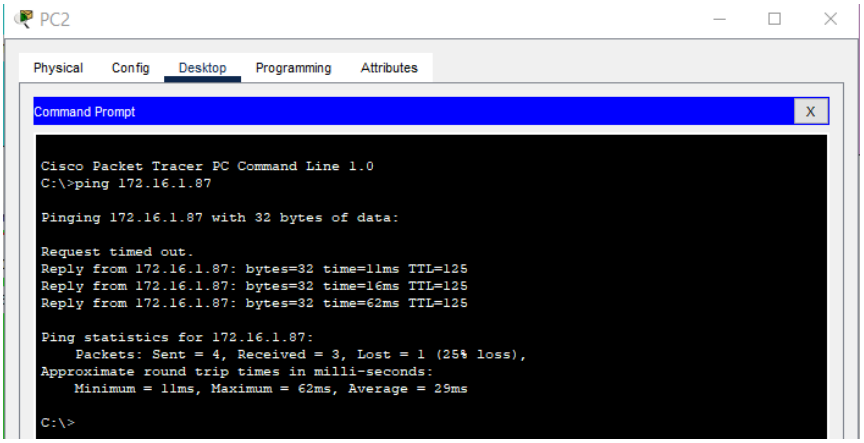
As we are getting a reply so our DHCP Protocol is setup correctly and is working good.

Ping from PC 0 (in LAN 1) to PC 4 (in LAN 3) :



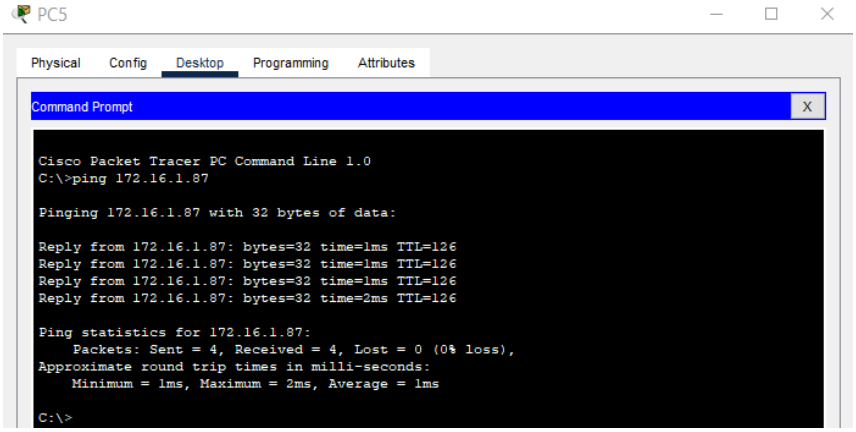
As we are getting a reply so our DHCP Protocol is setup correctly and is working good.

Ping from PC 2 (in LAN 2) to PC 6 (in LAN 4) :



As we are getting a reply so our DHCP Protocol is setup correctly and is working good.

Ping from PC 5 (in LAN 3) to PC 6 (in LAN 4) :

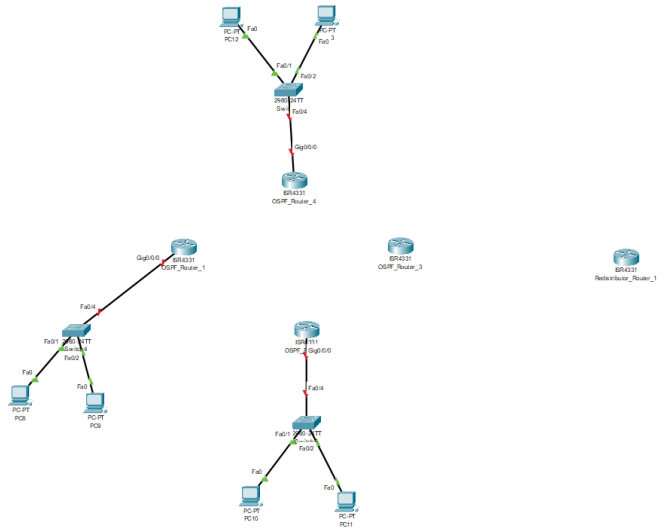


As we are getting a reply so our DHCP Protocol is setup correctly and is working good.

OSPF using DHCP server : (Stage 2)

Step 1 :

Place 4 routers, 3 switches and with each switch there are 2 PCs connected. 4 Routers are then connected with Redistributor Router to DHCP network server.

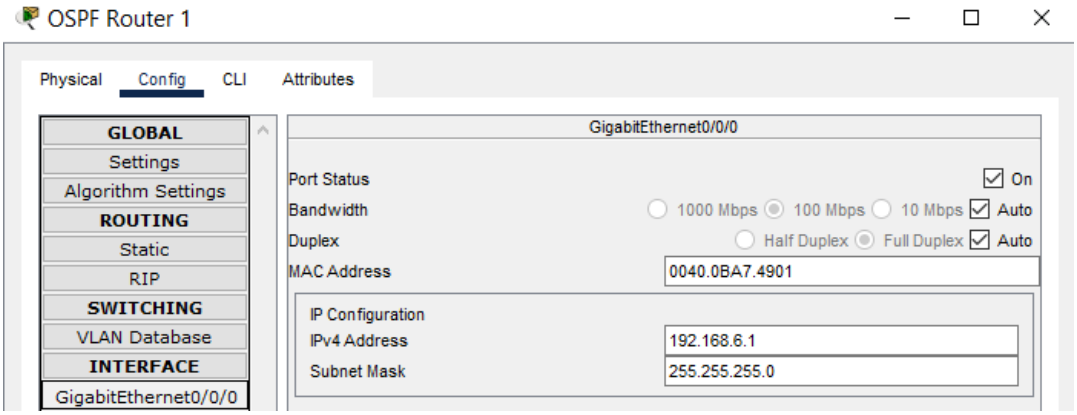


OSPF_LAN 1 :	OSPF_LAN 1 :
Default Gateway : 192.168.6.1	Starting I P s : 192.168.6.10 -
OSPF_LAN 2 :	OSPF_LAN 2 :
Default Gateway : 192.168.5.1	Starting I P s : 192.168.5.10 -
OSPF_LAN 3 :	OSPF_LAN 3 :
Default Gateway : 192.168.25.1	Starting I P s : 192.168.25.10

2. Go to each Router and in Gig write :

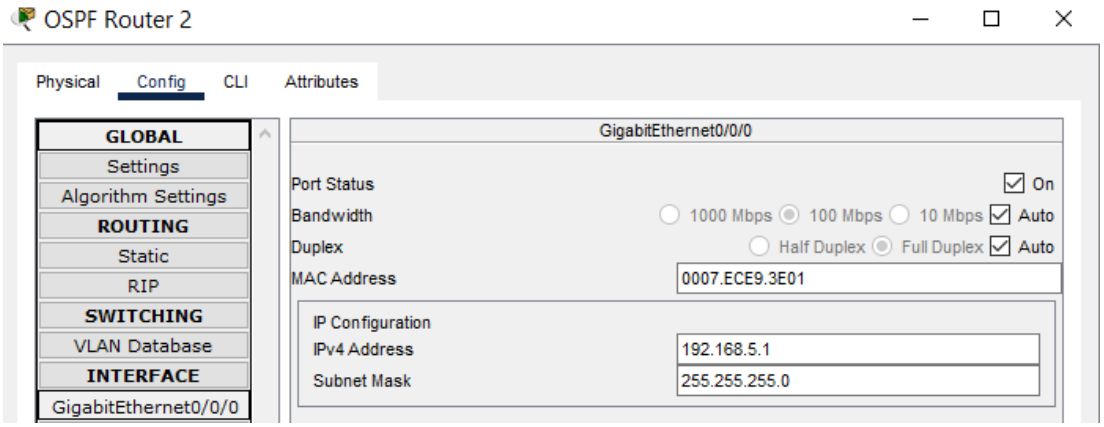
In OSPF-Router-1 :

In gig0/0/0 write the Default Gateway : 192.168.6.1 and turn it on.



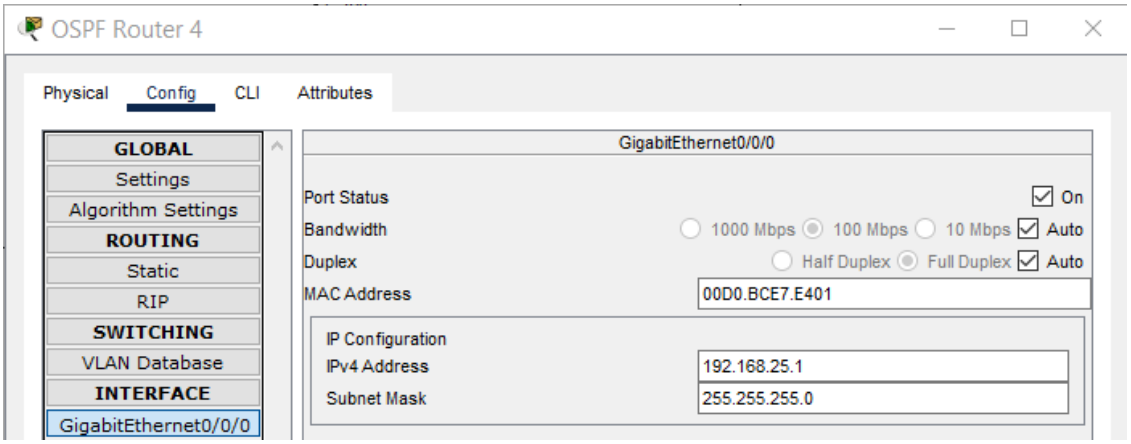
In OSPF-Router-2 :

In gig0/0/0 write the Default Gateway : 192.168.0.1 and turn it on.

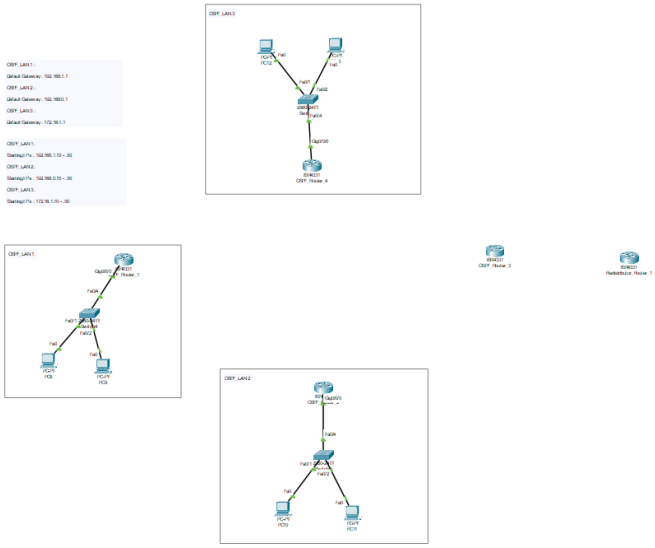


In OSPF-Router-4 :

In gig0/0/0 write the Default Gateway : 192.168.25.1 and turn it on.



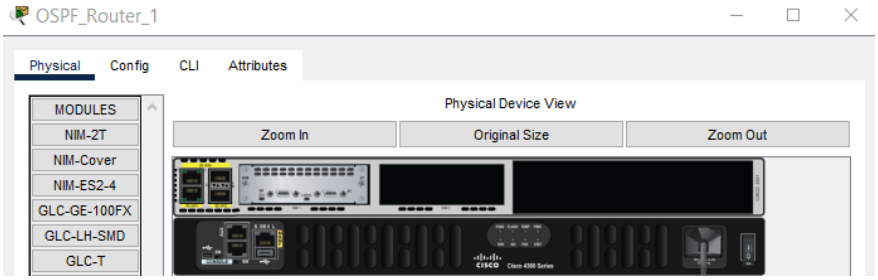
After this all the connected arrows will turn green.



3. Add Module in each Router :

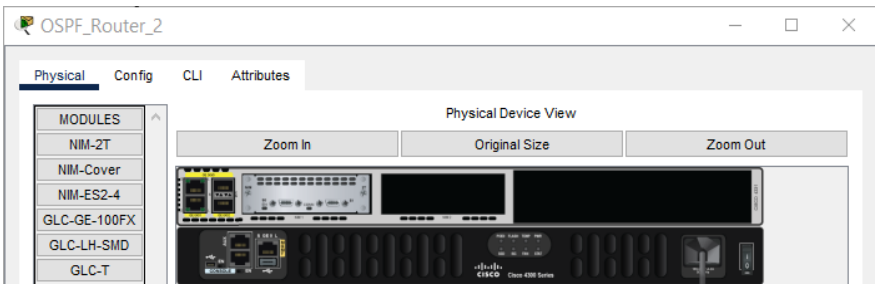
In OSPF-Router-1 :

Add 1 Module.



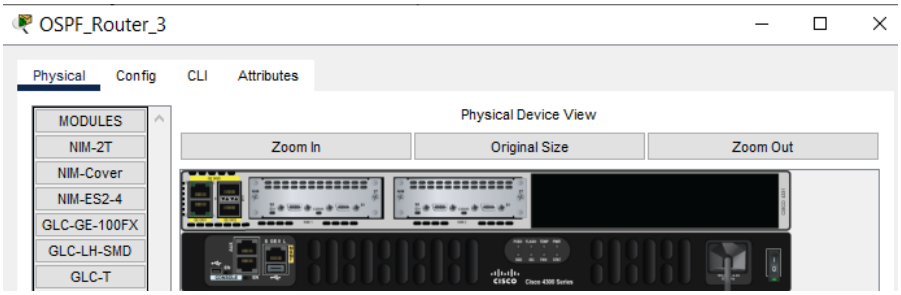
In OSPF-Router-2 :

Add 1 Module.



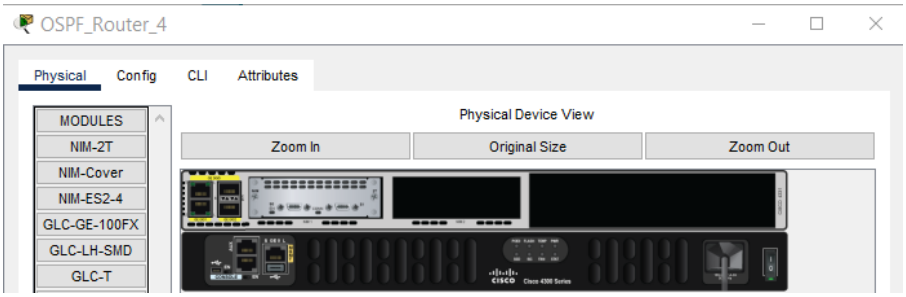
In OSPF-Router-3 :

Add 2 Module.



In OSPF-Router-4 :

Add 1 Module.

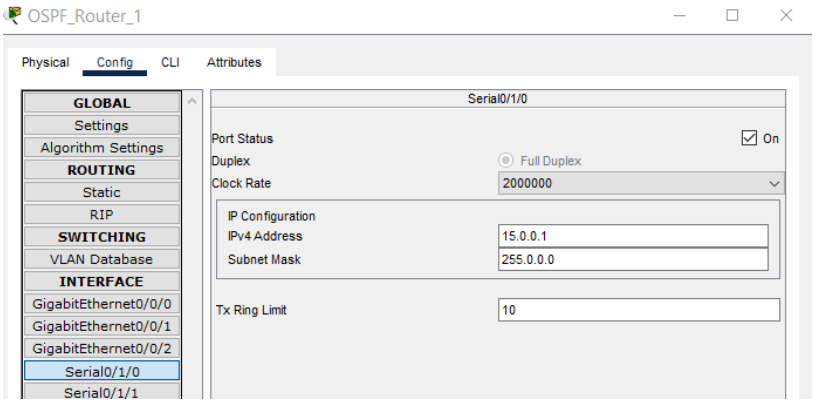


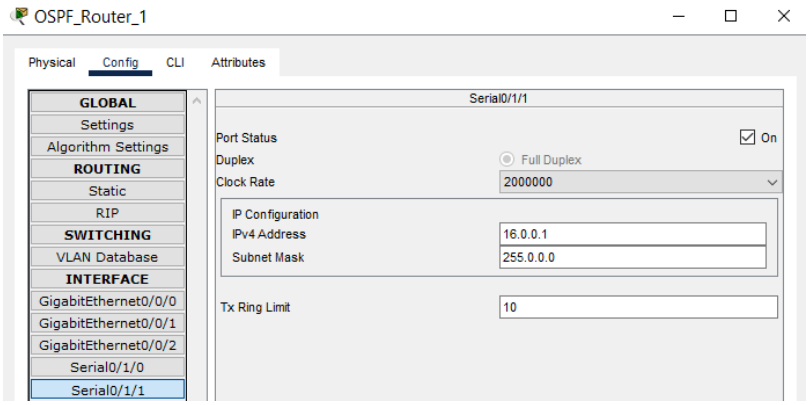
4. Assign IPs to serial ports :

In OSPF-Router-1 :

Give serial 0/1/0 15.0.0.1

Give serial 0/1/1 16.0.0.1 and turn them on.

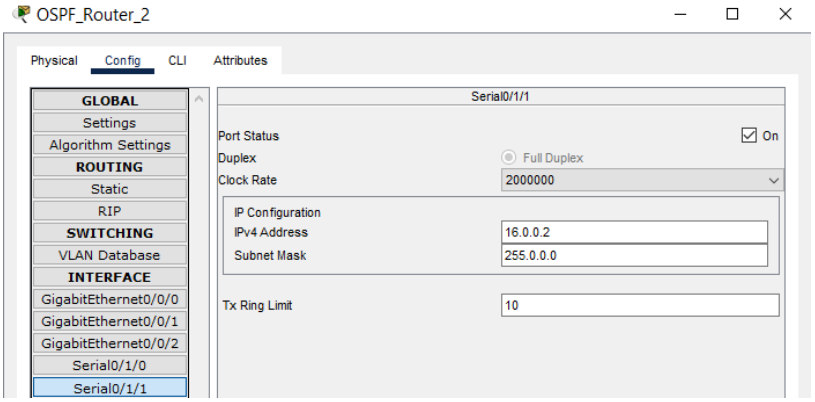


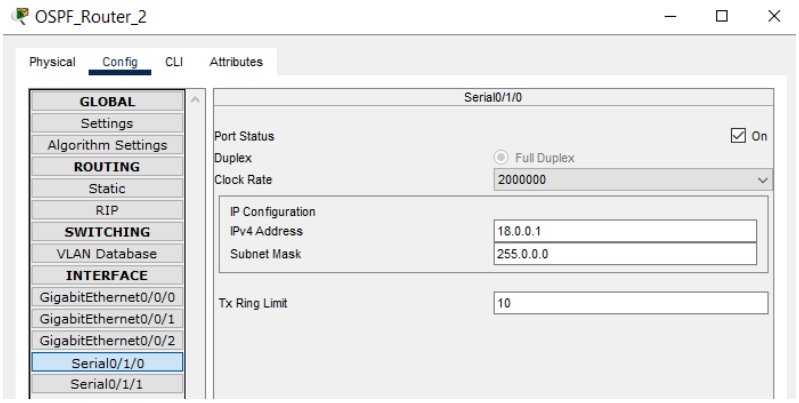


In OSPF-Router-2 :

Give serial 0/1/0 18.0.0.1

Give serial 0/1/1 16.0.0.2 and turn them on.



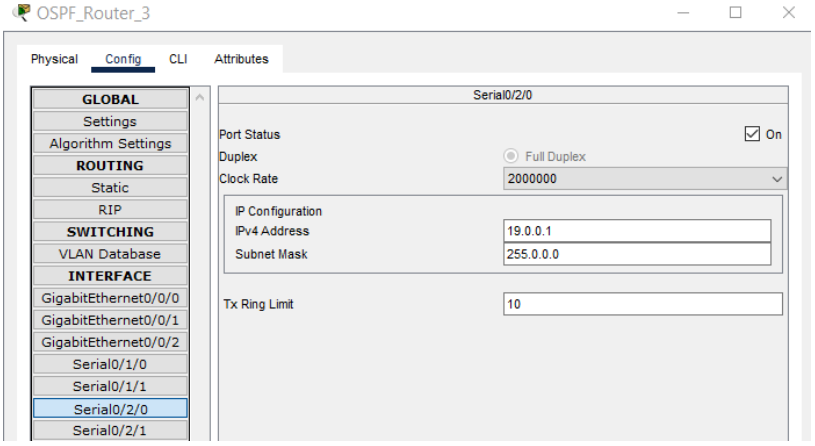


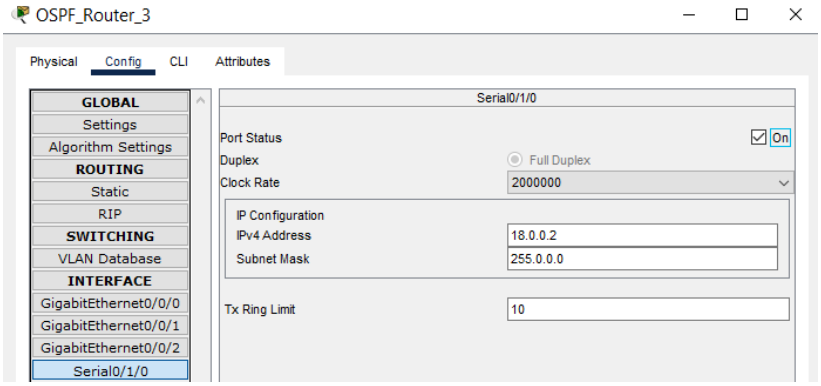
In OSPF-Router-3 :

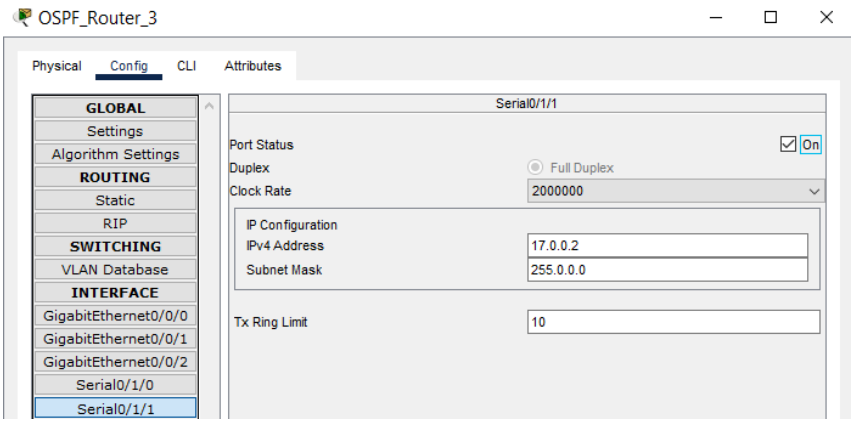
Give serial 0/1/1 17.0.0.2

Give serial 0/1/0 18.0.0.2

Give serial 0/2/0 19.0.0.1 and turn them on.



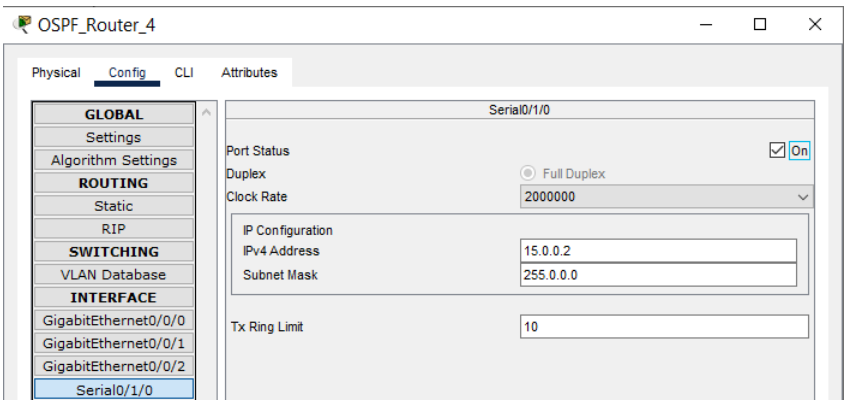


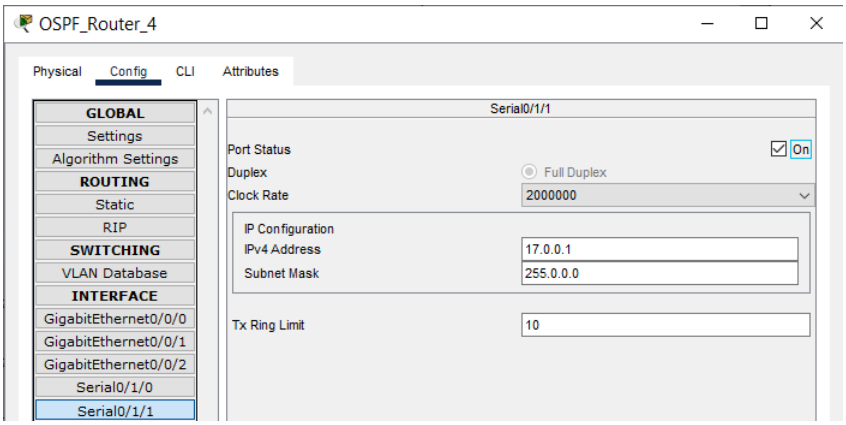


In OSPF-Router-4 :

Give serial 0/1/0 15.0.0.2

Give serial 0/1/1 17.0.0.1 and turn them on.

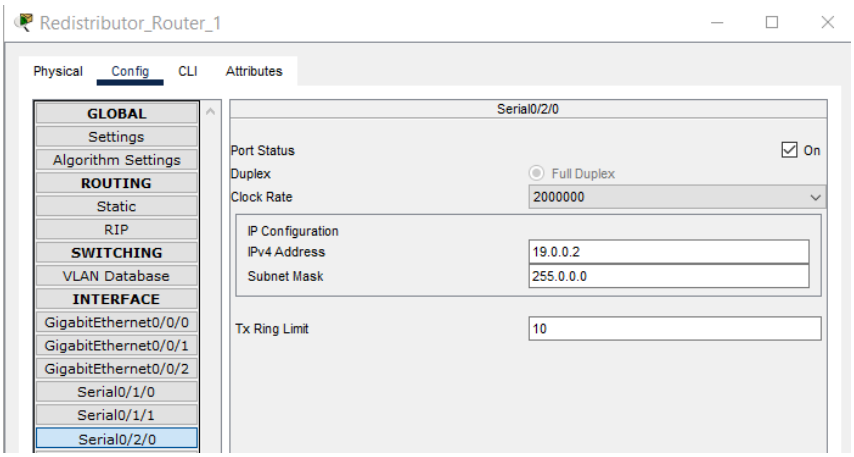


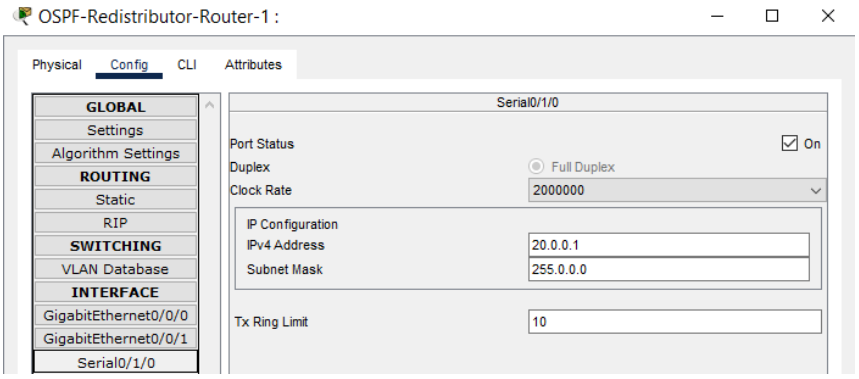


In OSPF-Redistributor-Router-1 :

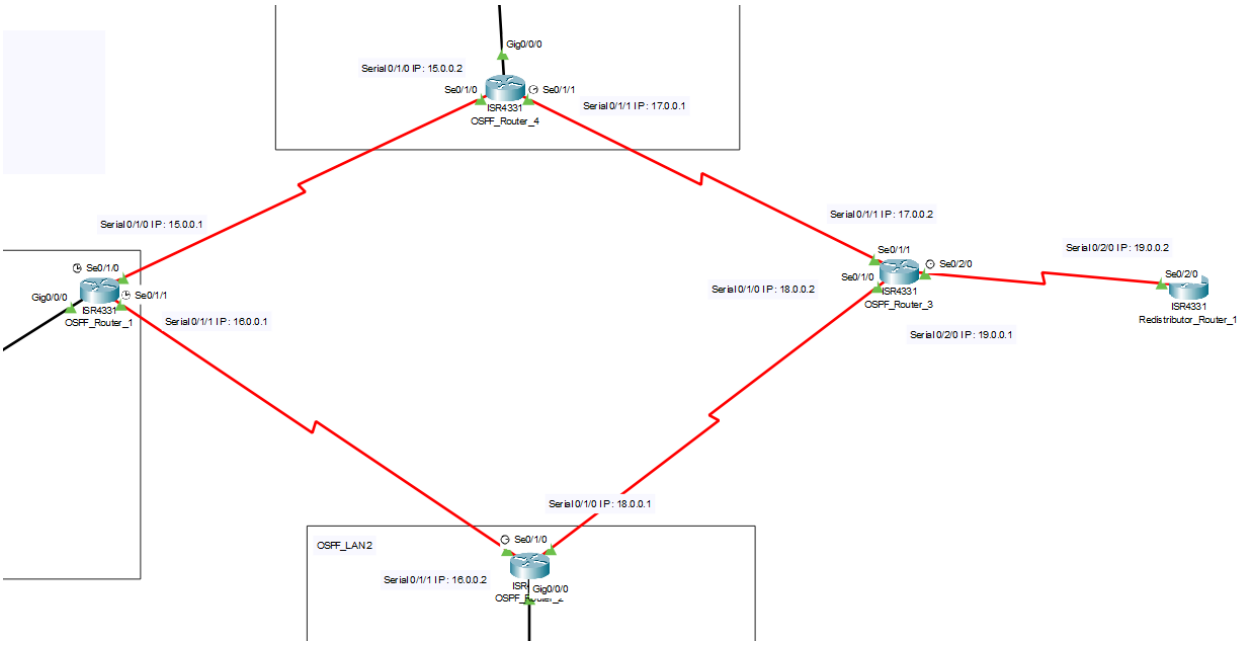
Give serial 0/2/0 19.0.0.2

Give serial 0/1/0 20.0.0.1





After that all arrows in connections of wires will be turned green.



Wildcard Calculation :

	Default Gateway	Subnet Mask :	Wildcard Maks:
LAN 1 :	192.168.6.1	255.255.255.0	0.0.0.255
LAN 2 :	192.168.5.1	255.255.255.0	0.0.0.255
LAN 3 :	192.168.25.1	255.255.255.0	0.0.0.255

Other Networks :

Network	Wildcard Mask:
15.0.0.0	0.255.255.255
16.0.0.0	0.255.255.255
17.0.0.0	0.255.255.255
18.0.0.0	0.255.255.255
19.0.0.0	0.255.255.255

Step 2 : Go to OSPF-Router-1 on CLI write :

en

conf t

router ospf 1

network 192.168.6.0 0.0.0.255 area 0

network 15.0.0.0 0.255.255.255 area 0

network 16.0.0.0 0.255.255.255 area 0

```
Router>en
Router#
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#router ospf 1
Router(config-router)#network 192.168.6.0 0.255.255.255 area 0
Router(config-router)#network 16.0.0.0 0.0.0.255 area 0
Router(config-router)#network 15.0.0.0 0.0.0.255 area 0
Router(config-router)#
Router(config-router)#
Router(config-router)#
Router#
```

On OSPF-Router-2 (on CLI) :

en

conf t

router ospf 1

network 192.168.5.0 0.0.0.255 area 0

network 16.0.0.0 0.255.255.255 area 0

network 18.0.0.0 0.255.255.255 area 0

```
Router>en
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#
Router(config)#router ospf 1
Router(config-router)#network 16.0.0.0 0.255.255.255 area 0
Router(config-router)#network 18.0.0.0 0.255.255.255 area 0
Router(config-router)#network 192.168.5.0 0.0.0.255 area 0
Router(config-router)#
```

On OSPF-Router-3 (on CLI) :

en

conf t

router ospf 1

network 17.0.0.0 0.255.255.255 area 0

network 18.0.0.0 0.255.255.255 area 0

network 19.0.0.0 0.255.255.255 area 0

```
Router>en
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#router ospf 1
Router(config-router)#network 17.0.0.0 0.255.255.255 area 0
Router(config-router)#netwok 18.0.0.0 0.255.255.255 area 0
                         ^
% Invalid input detected at '^' marker.

Router(config-router)#network 18.0.0.0 0.255.255.255 area 0
Router(config-router)#network
02:19:12: %OSPF-5-ADJCHG: Process 1, Nbr 192.168.0.1 on Serial0/1/0 from LOADING to FULL,
Loading Done

% Incomplete command.
Router(config-router)#network 19.0.0.0 0.255.255.255 area 0
```

On OSPF-Router-4 (on CLI) :

en

conf t

router ospf 1

network 192.168.25.0 0.0.0.255 area 0

network 15.0.0.0 0.255.255.255 area 0

network 17.0.0.0 0.255.255.255 area 0

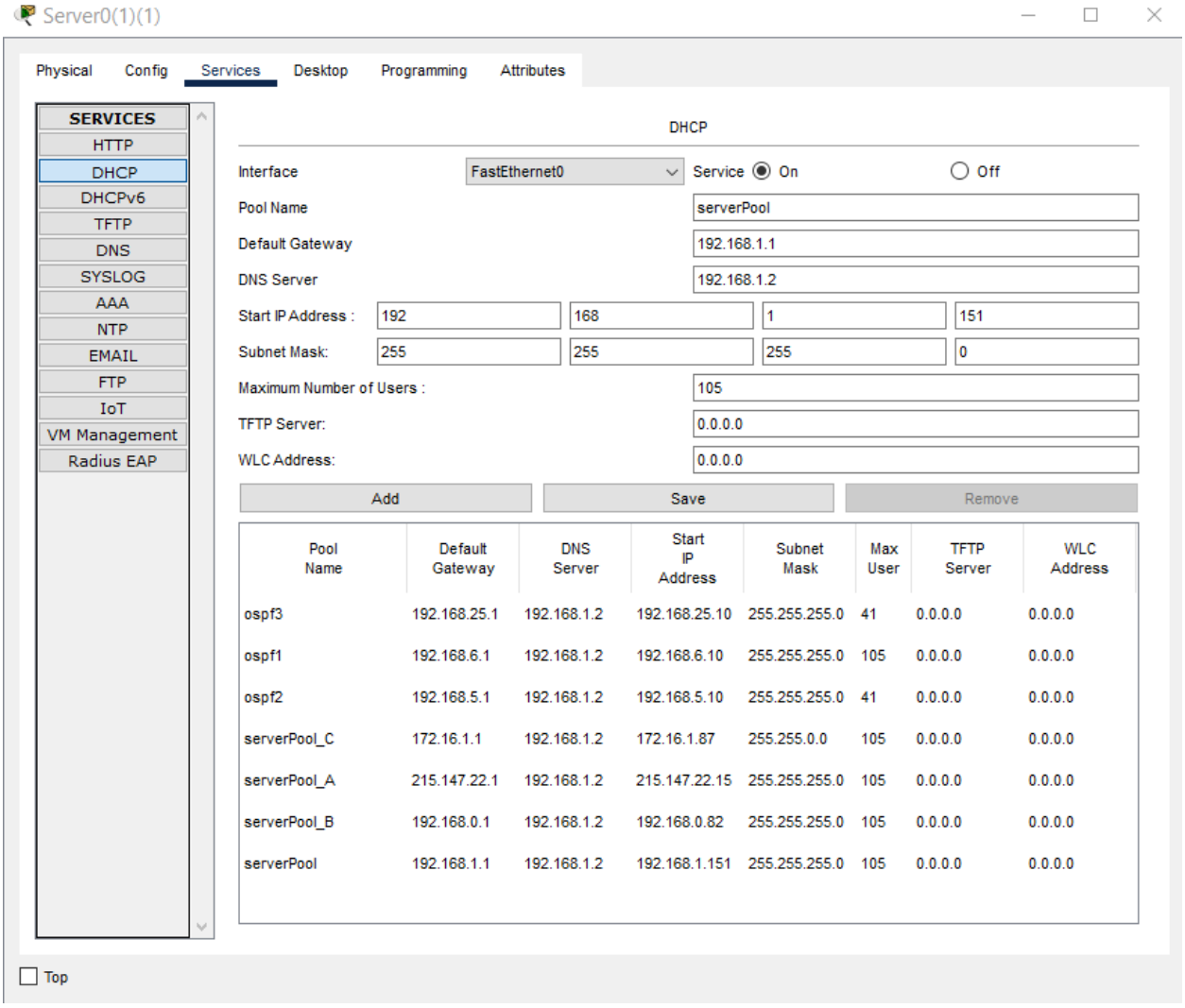
```
Router>en
Router#
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#
Router(config)# router ospf 1
Router(config-router)#network 17.0.0.0 0.255.255.255 area 0
Router(config-router)#network 15.0.0.0 0.255.255.255 area 0
Router(config-router)#network 192.168.25.0 0.0.0.255 area 0
```

Step 3 : Go to DHCP and write Poolnames :

Write the Poolname as “ospf1” , Default Gateway : 192.168.6.1 & Starting IP : 192.168.6.10

Write the Poolname as “ospf2” , Default Gateway : 192.168.5.1 & Starting IP : 192.168.5.10

Write the Poolname as “ospf3” , Default Gateway : 192.168.25.1 & Starting IP : 192.168.25.10



Pool Name	Default Gateway	DNS Server	Start IP Address	Subnet Mask	Max User	TFTP Server	WLC Address
ospf3	192.168.25.1	192.168.1.2	192.168.25.10	255.255.255.0	41	0.0.0.0	0.0.0.0
ospf1	192.168.6.1	192.168.1.2	192.168.6.10	255.255.255.0	105	0.0.0.0	0.0.0.0
ospf2	192.168.5.1	192.168.1.2	192.168.5.10	255.255.255.0	41	0.0.0.0	0.0.0.0
serverPool_C	172.16.1.1	192.168.1.2	172.16.1.87	255.255.0.0	105	0.0.0.0	0.0.0.0
serverPool_A	215.147.22.1	192.168.1.2	215.147.22.15	255.255.255.0	105	0.0.0.0	0.0.0.0
serverPool_B	192.168.0.1	192.168.1.2	192.168.0.82	255.255.255.0	105	0.0.0.0	0.0.0.0
serverPool	192.168.1.1	192.168.1.2	192.168.1.151	255.255.255.0	105	0.0.0.0	0.0.0.0

Step 4 : On each Router other than Redistributor Router : (In CLI write)

en

conf t

int gig0/0/0

ip helper-address 192.168.1.3

exit

exit

This will ask the DHCP server whose address is 192.168.1.3 to assign IPs to PCs.

```
Router>en
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#int gig0/0/0
Router(config-if)#ip helper-address 192.168.1.3
Router(config-if)#exit
Router(config)#exit
```

```
Router>en
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#int gig0/0/0
Router(config-if)#ip helper-address 192.168.1.3
Router(config-if)#exit
Router(config)#exit
```

```
Router>en
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#int gig0/0/0
Router(config-if)#ip helper-address 192.168.1.3
Router(config-if)#exit
Router(config)#exit
```

Step 5 : On Redistributor Router : (In CLI write)

en

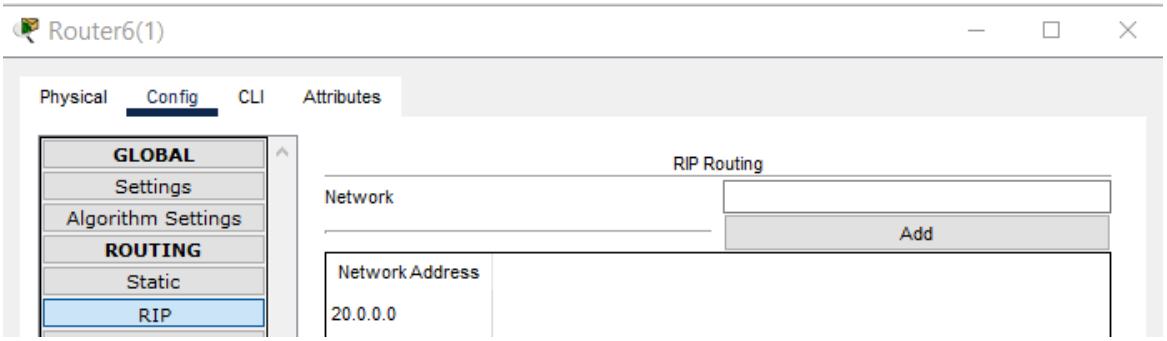
conf t

router ospf 2

network 19.0.0.0 0.0.0.255 area 0

```
Router#en
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#
Router(config)#router ospf 1
Router(config-router)#network 19.0.0.0 0.0.0.255 area 0
Router(config-router)#
00:14:28: %OSPF-5-ADJCHG: Process 1, Nbr 19.0.0.1 on Serial0/2/0 from LOADING to FULL,
Loading Done
```

Also, add RIP from Redistributor side to RIP side.



Now, On Redistributor Router (for Redistribution between OSPF and RIP in CLI) write :

en

conf t

router rip

redistribute ospf 1 metric 2

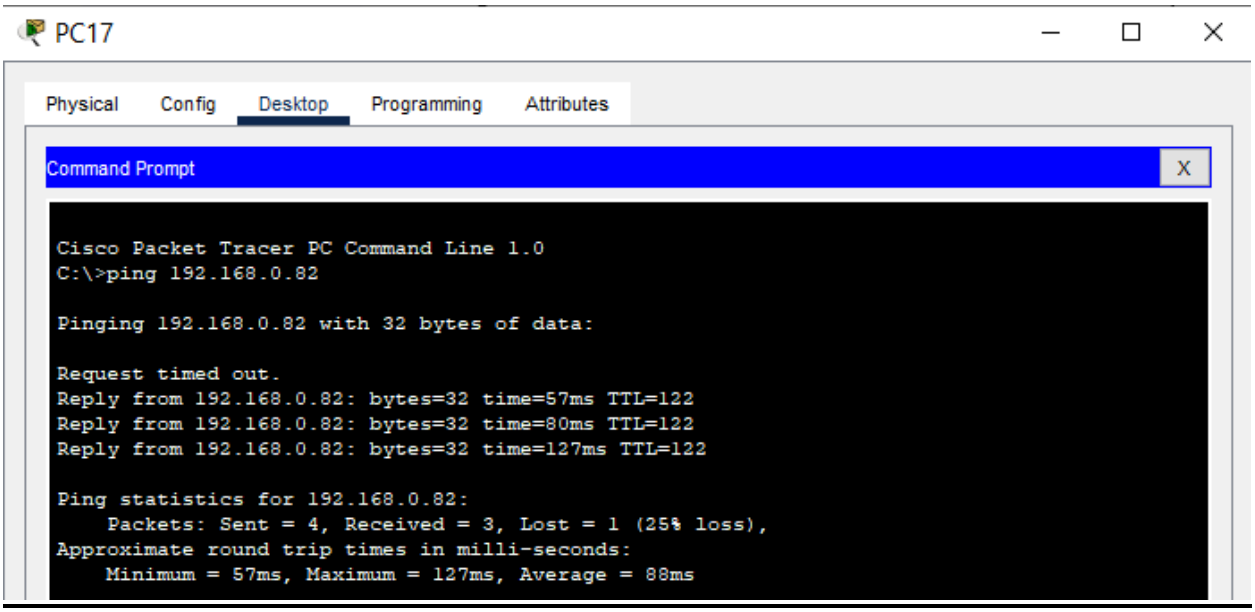
router ospf 1

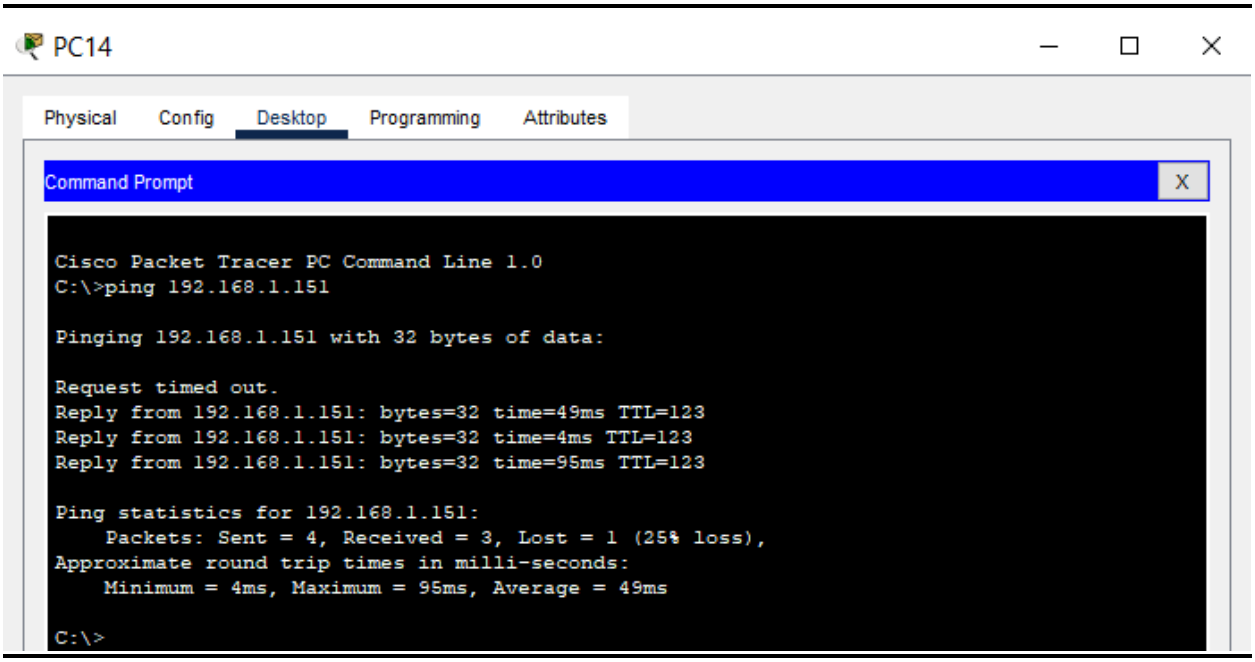
redistribute rip metric 2

```
Router>en
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#router rip
Router(config-router)#red
Router(config-router)#redistribute ospf 1 metric 2
Router(config-router)#reouter ospf 1
                     ^
% Invalid input detected at '^' marker.

Router(config-router)#router ospf 1
Router(config-router)#red
Router(config-router)#redistribute rip metric 2
% Only classful networks will be redistributed
Router(config-router)#
```

After that when we ping from different PCs in OSPF network to PCs in RIP network :



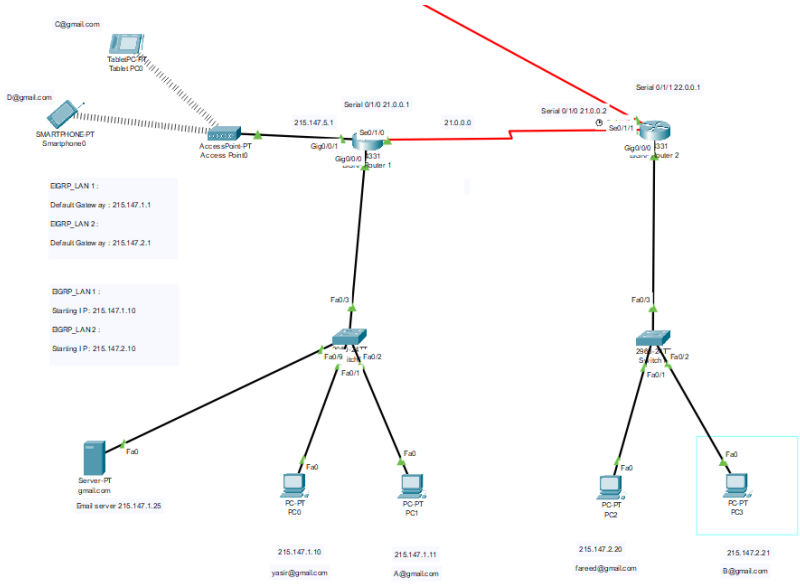


That means our RIP, OSPF and Redistribution between them is working properly and is good.

EIGRP using DHCP server : (Stage 3)

Step 1 :

Place 2 routers, 2 switches and with each switch there are 2 PCs connected. 2 Routers are then connected with Redistributor Router to OSPF & RIP (DHCP) network server.

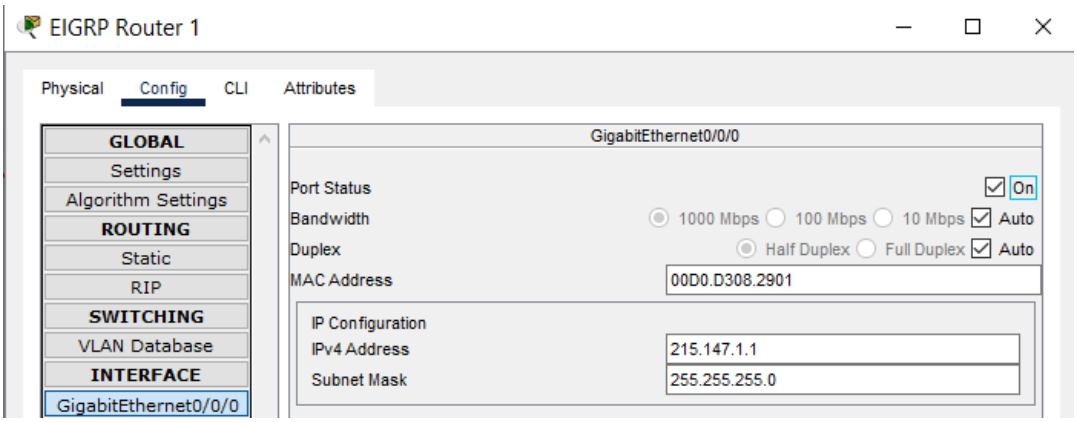


EIGRP_LAN 1 :	EIGRP_LAN 1 :
Starting I P : 215.147.1.10	Default Gateway : 215.147.1.1
EIGRP_LAN 2 :	EIGRP_LAN 2 :
Starting I P : 215.147.2.10	Default Gateway : 215.147.2.1

2. Go to each Router and in Gig write :

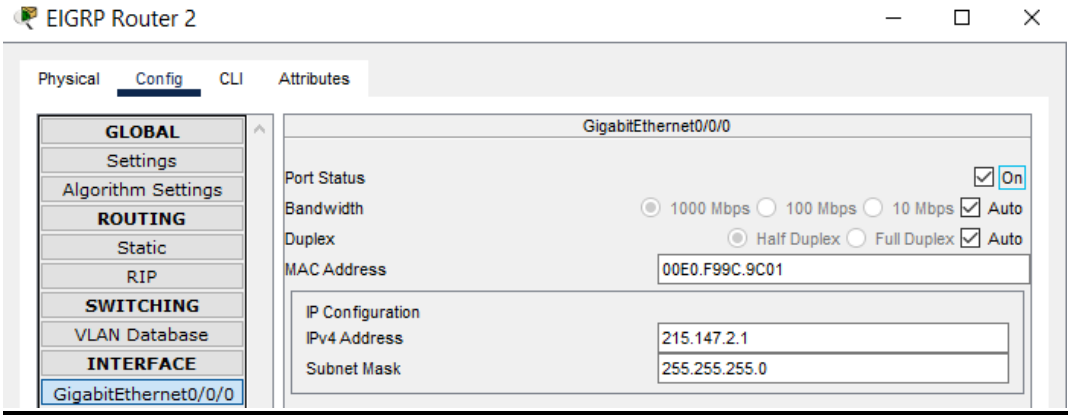
EIGRP Router 1 :

In gig0/0/0 write the Default Gateway : 215.147.1.1 and turn it on.

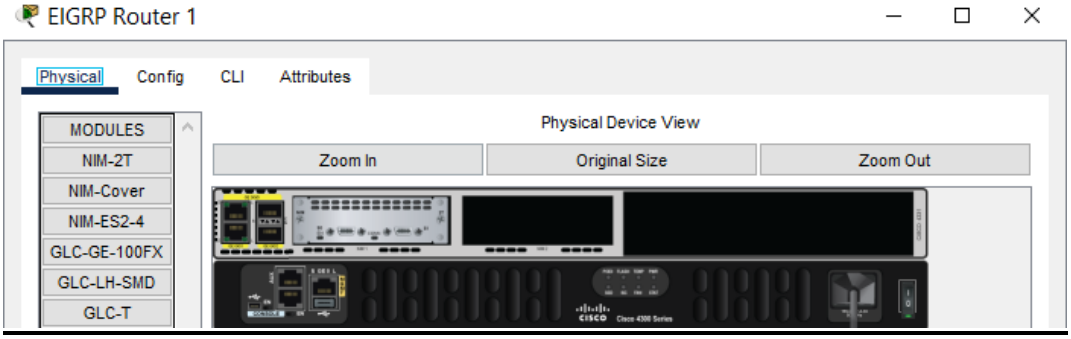


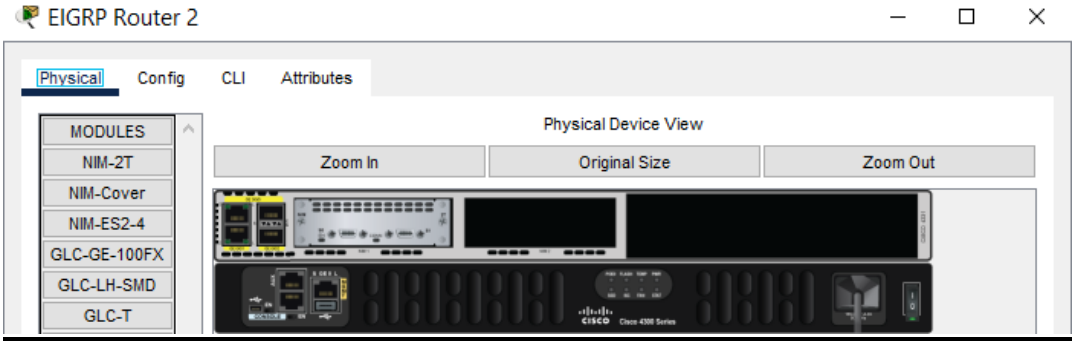
EIGRP Router 2 :

In gig0/0/0 write the Default Gateway : 215.147.2.1 and turn it on.



After that add modules in Routers and connect routers with each other using serial ports.

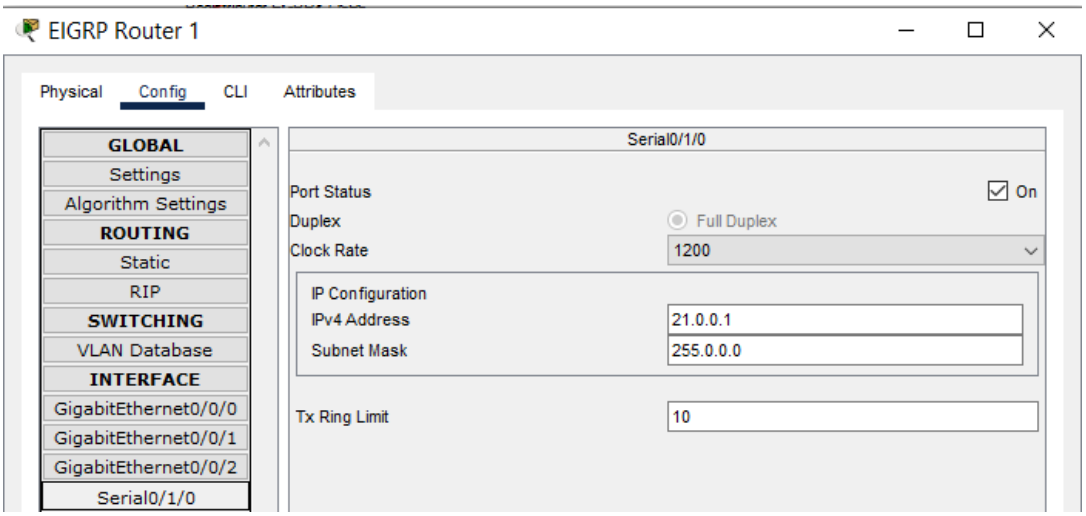




3. Give IP addresses to serial ports :

EIGRP Router 1 :

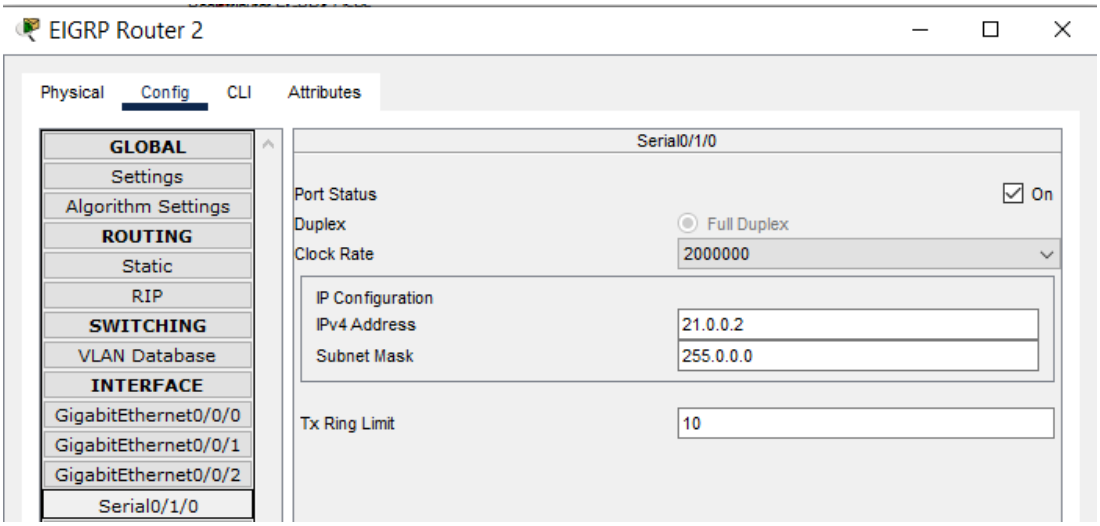
Give serial 0/1/0 IP address : 21.0.0.1

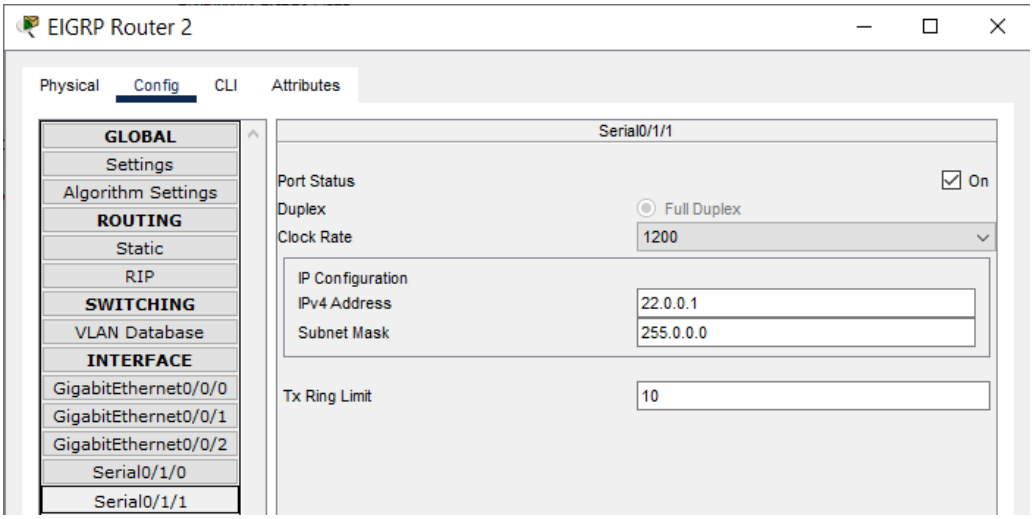


In EIGRP Router 2 :

Give serial 0/1/0 IP address 21.0.0.2

Give serial 0/1/1 IP address 22.0.0.1

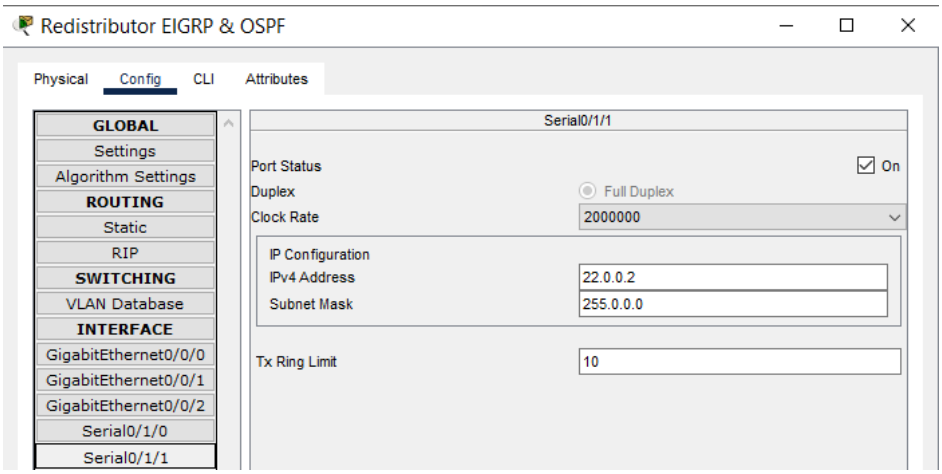


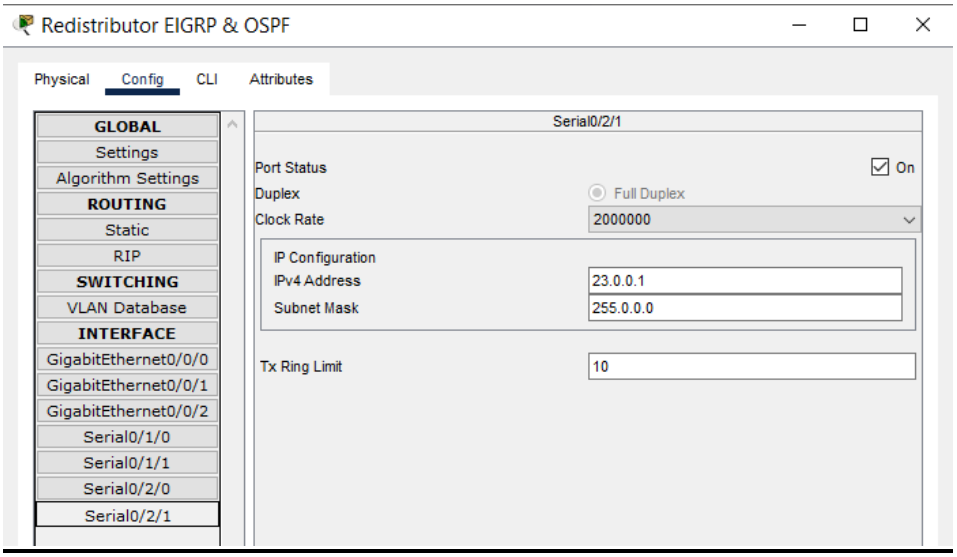


On Redistributor Router (EIGRP to OSPF):

Give serial 0/1/1 IP address 22.0.0.2

Give serial 0/2/1 IP address 23.0.0.1





Step 2 : Go to EIGRP-Router-1 on CLI write

en

conf t

router eigrp 10

network 215.147.1.0

network 21.0.0.0

```
Router>en
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#router eigrp 10
Router(config-router)#network 215.147.1.0
Router(config-router)#network 21.0.0.0
Router(config-router)#
```

2. Go to EIGRP-Router-2 on CLI write.

en

conf t

router eigrp 10

network 215.147.2.0

network 21.0.0.0

network 23.0.0.0

```
Router#en
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#router eigrp 10
Router(config-router)#network 21.0.0.0
Router(config-router)#
%DUAL-5-NBRCHANGE: IP-EIGRP 10: Neighbor 21.0.0.1 (Serial0/1/0) is up: new adjacency
network 215.147.2.0
Router(config-router)#network 215.147.2.0
Router(config-router)#network 23.0.0.0
Router(config-router)#
```

3. Go to Redistributor EIGRP & OSPF Router on CLI write.

For EIGRP Side:

en

conf t

router eigrp 10

network 22.0.0.0

```
Router>
Router>en
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#router eigrp 10
Router(config-router)#network 22.0.0.0
Router(config-router)#
```

For OSPF Side:

en

conf t

router ospf 1

network 23.0.0.0 0.0.0.255 area 0

```
Router#en
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#
Router(config)#router ospf 1
Router(config-router)#network 23.0.0.0 0.0.0.255 area 0
Router(config-router)#exit
Router(config)#exit
Router#
```

Step 3 : Go to EIGRP-Router-1 on CLI write

en

conf t

int gig0/0/0

ip helper-address 192.168.1.3

exit

exit

```
Router#en
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#int gig0/0/0
Router(config-if)#ip helper-address 192.168.1.3
Router(config-if)#
Router(config-if)#
Router(config-if)#
Router(config-if)#
```

It will ask 192.168.1.3 (server) to assign it IP address automatically.

2. Go to EIGRP-Router-2 on CLI write

en

conf t

int gig0/0/0

ip helper-address 192.168.1.3

exit

exit

```
Router#en
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#int gig0/0/0
Router(config-if)#ip helper address 192.168.1.3
                                   ^
% Invalid input detected at '^' marker.

Router(config-if)#ip helper address 192.168.1.3
                                   ^
% Invalid input detected at '^' marker.

Router(config-if)#ip helper-address 192.168.1.3
```

Step 4 : Go to Redistributor EIGRP & OSPF Router on CLI write.

en

conf t

router ospf 1

redistribute eigrp 10 subnets

router eigrp 10

redistribute ospf 1 metric 2

redistribute ospf 1 metric 2 2 255 255 2 (Where **2** is Bandwidth in kbps

2 is delay in tens of milliseconds

255 is reliability 255 means 100% reliable

255 is load 1 is low while 255 is high load

2 is MTU Maximum Transmission Unit)

```
Router#en
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#router ospf 1
Router(config-router)#red
Router(config-router)#redistribute eigrp 10 subnets
Router(config-router)#red
Router(config-router)#redistribute eigrp 10
% Only classful networks will be redistributed
```

```
Router#
Router#
Router#en
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#router eigrp 10
Router(config-router)#redistribute ospf 1 metric 2 2 255 255 2
Router(config-router)#
```

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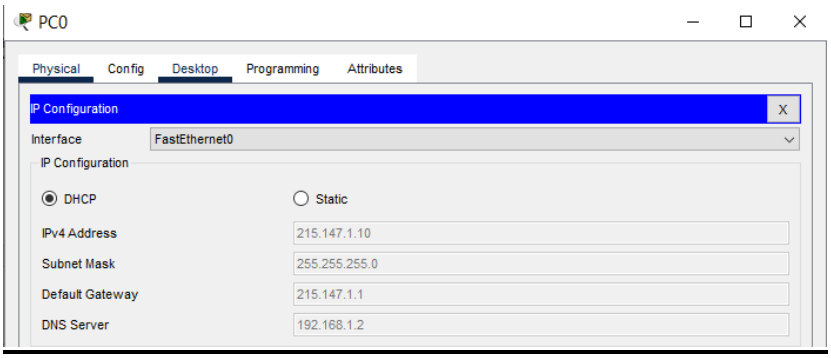
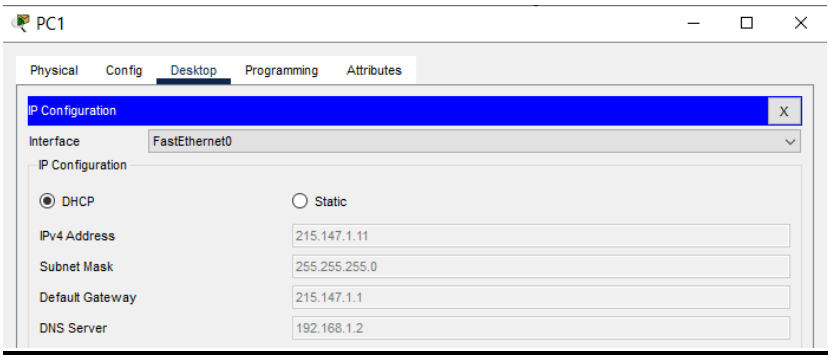
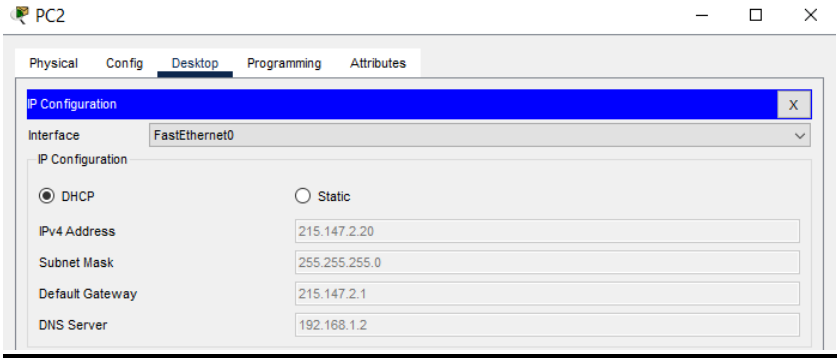
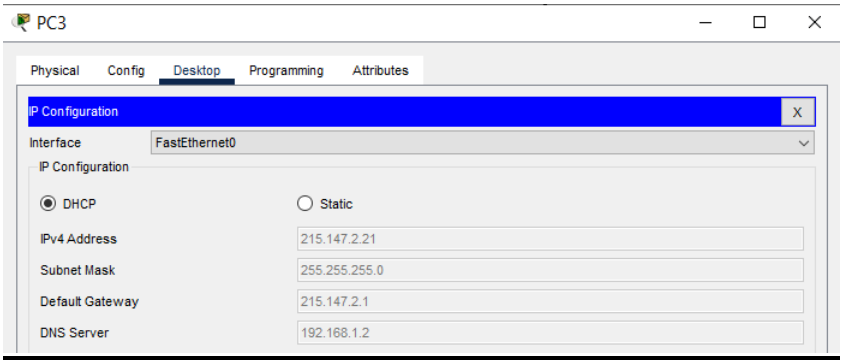
Step 5 : After that goto each PC in EIGRP (Desktop->IP Configuration, Click on DHCP) We can see that DHCP server automatically assign IP addresses to PCs .



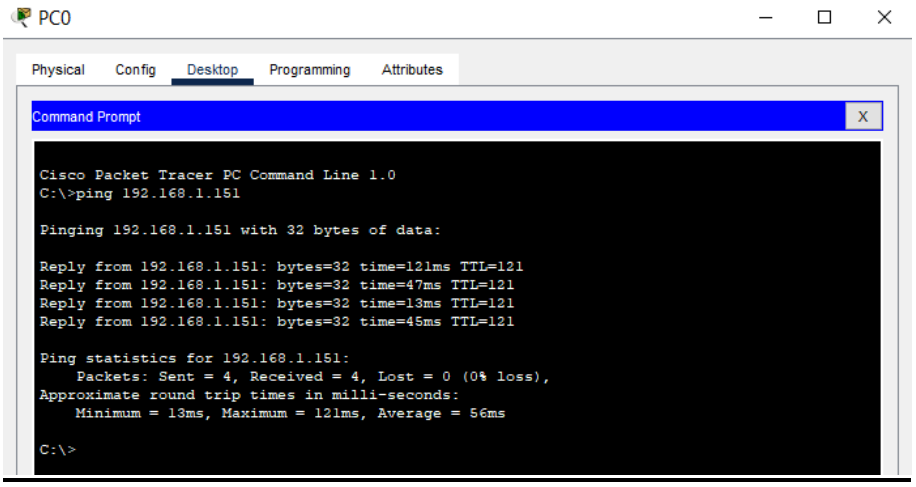




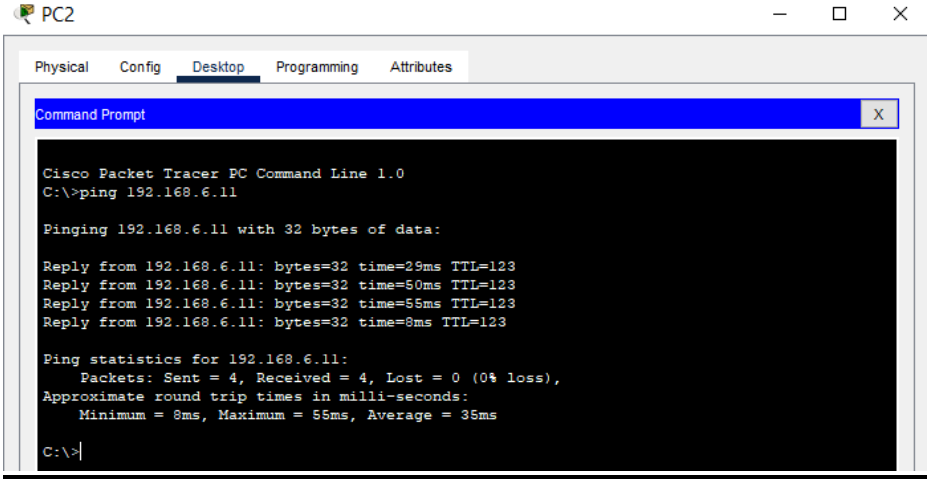


Step 6 : Ping any PC in OSPF and RIP to check our Redistribution is working correctly.

When we ping from PC0 in EIGRP to PC0 in RIP.



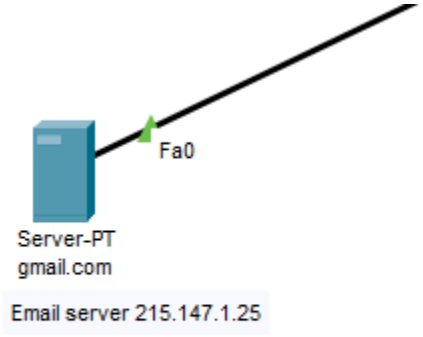
When we ping from PC2 in EIGRP to PC17 in OSPF.



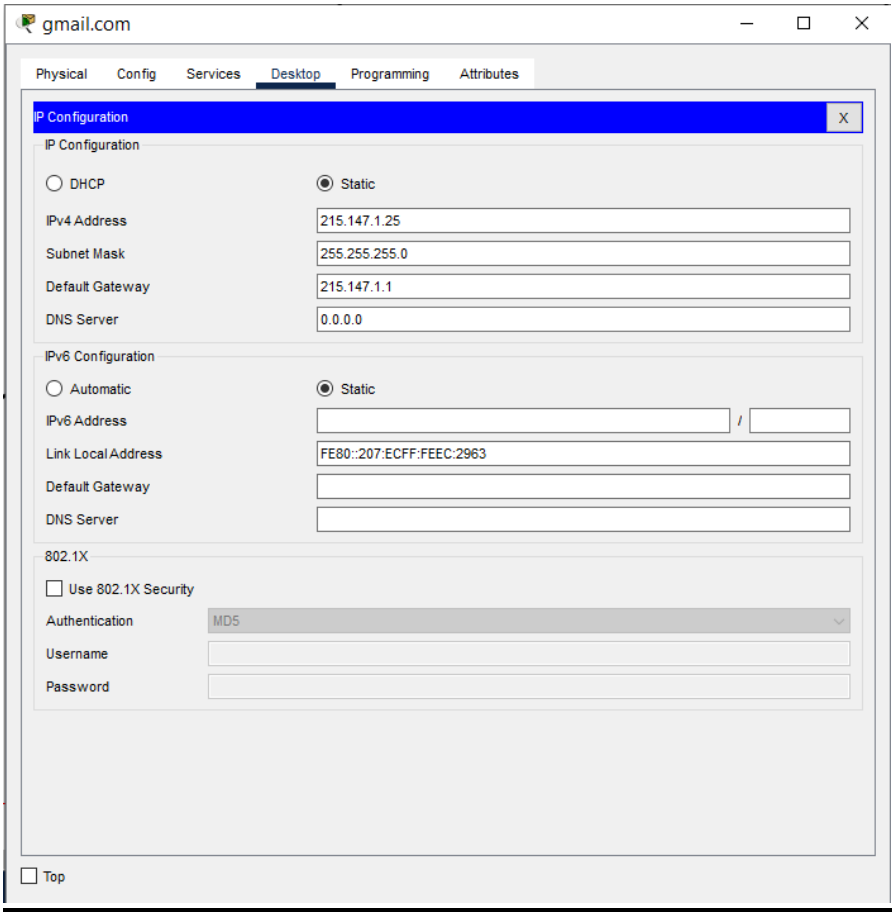
That means our EIGRP is working perfectly fine along with Redistribution between OSPF and RIP.

Email Configuration using server in EIGRP : (Stage 4)

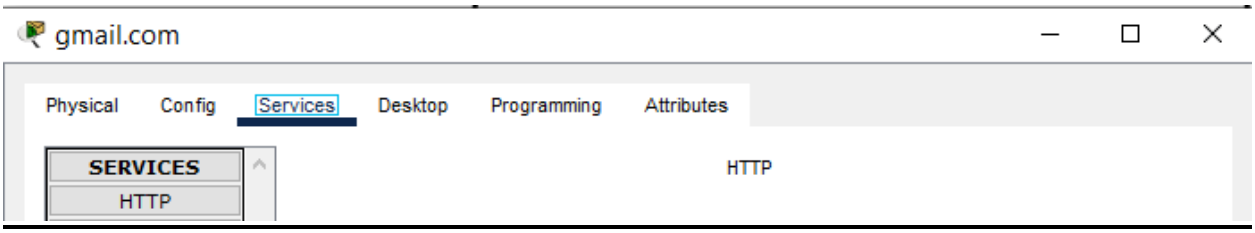
Step 1 : Attach a server in EIGRP topology.

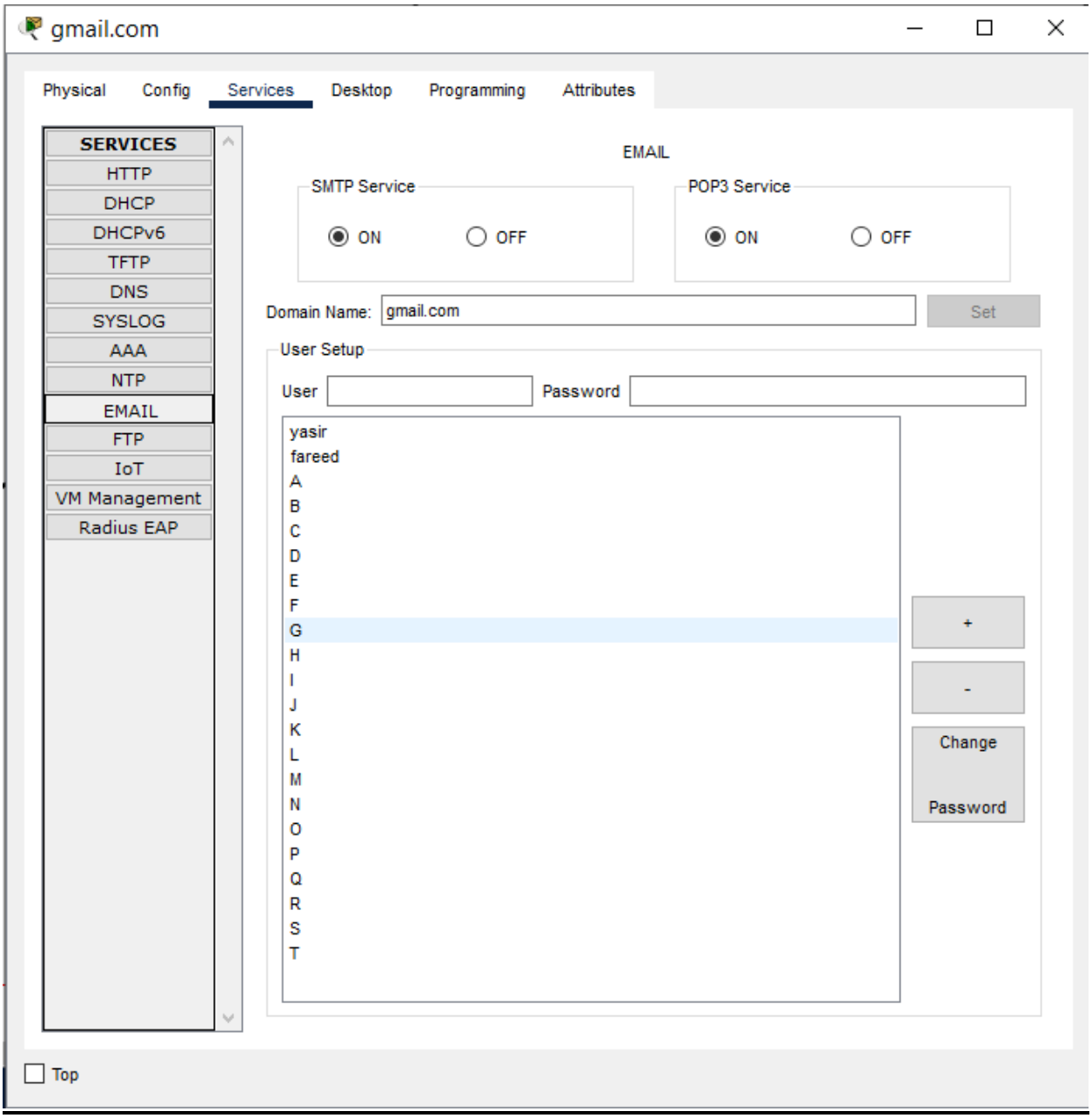


2. Configure the Email server write its IP address and Default Gateway.

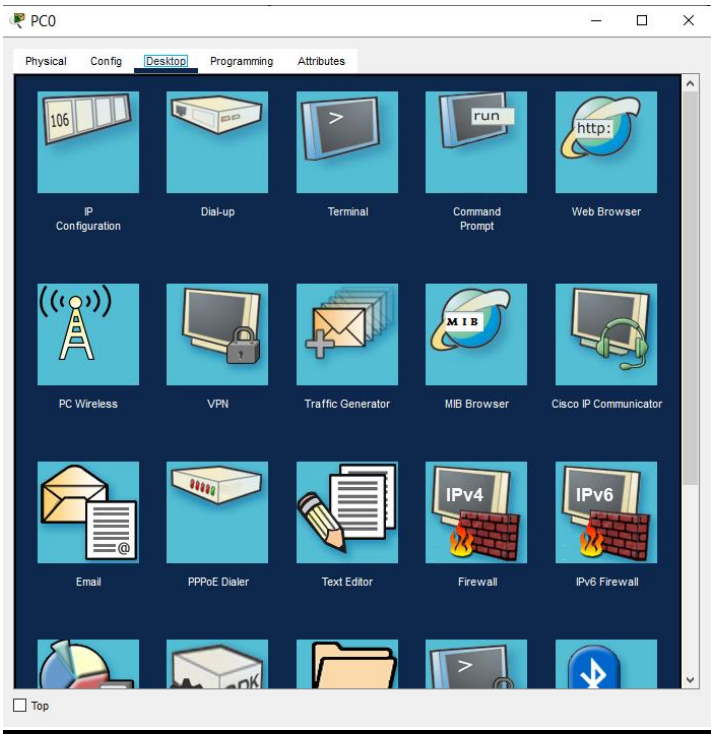


Step 2 : On that server in services goto Email. Add Domain Name and Username and Password that we want to assign to each PC(device) in network.

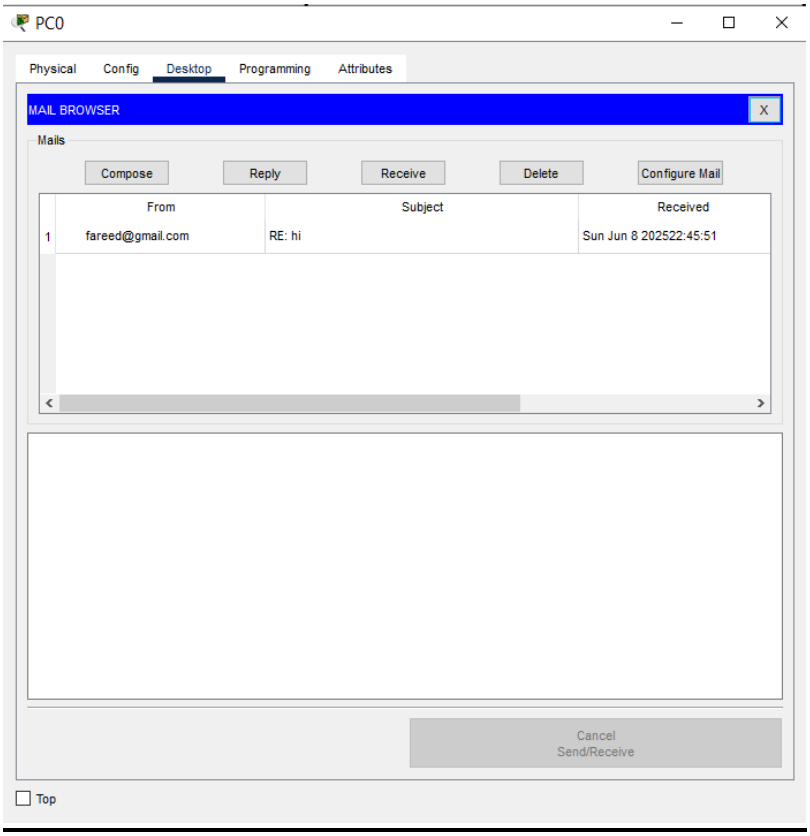




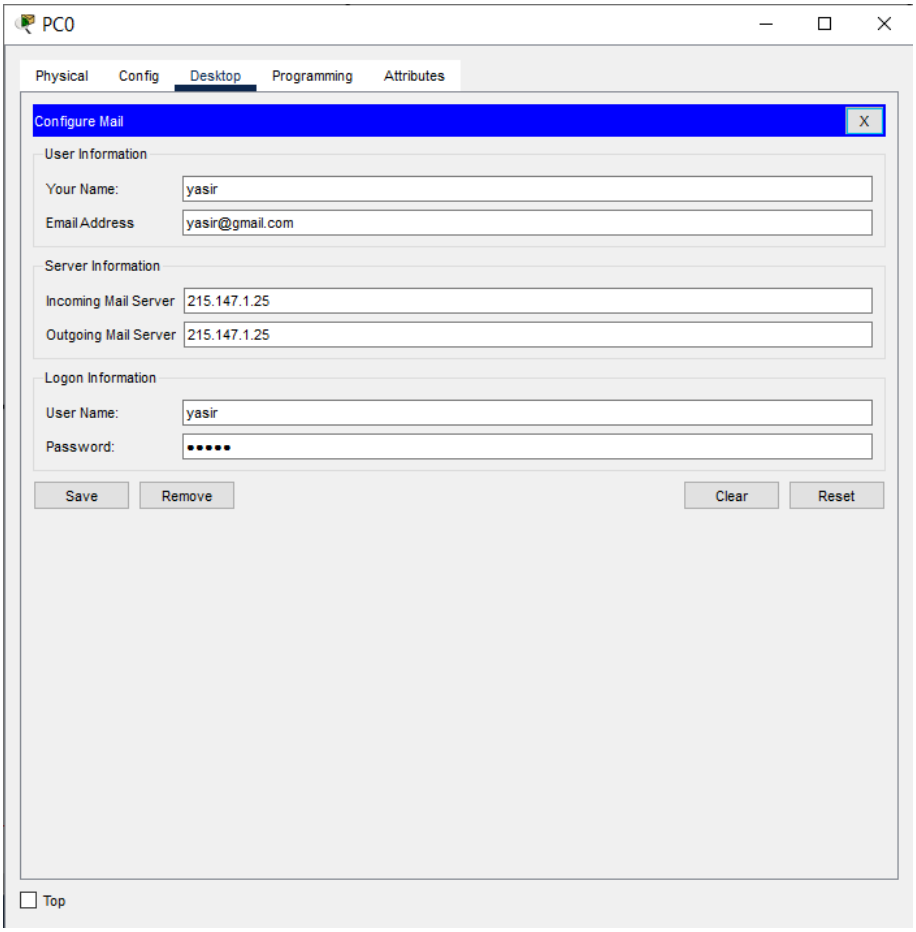
2. Click on any PC in Desktop click on Email.



3. Click on Configure mail



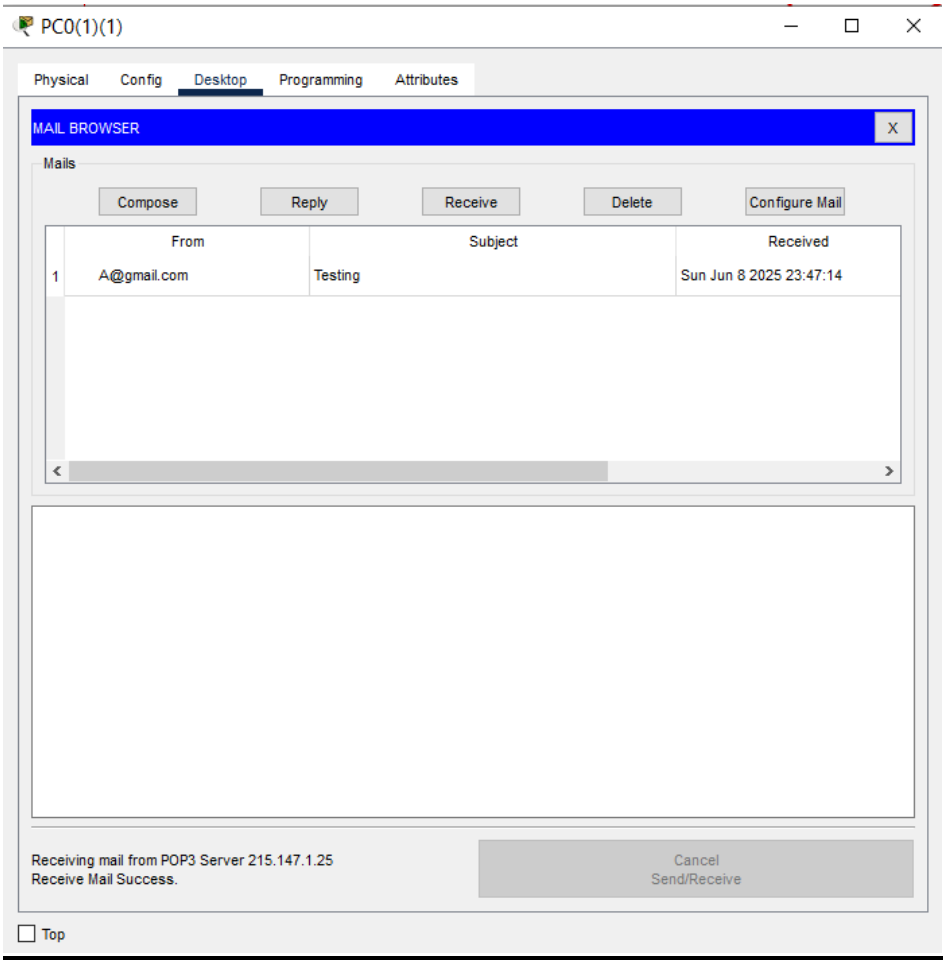
Step 3 : Write the username, EmailAddress, Incoming + outgoing Mail server IP address and also write the Logon information.

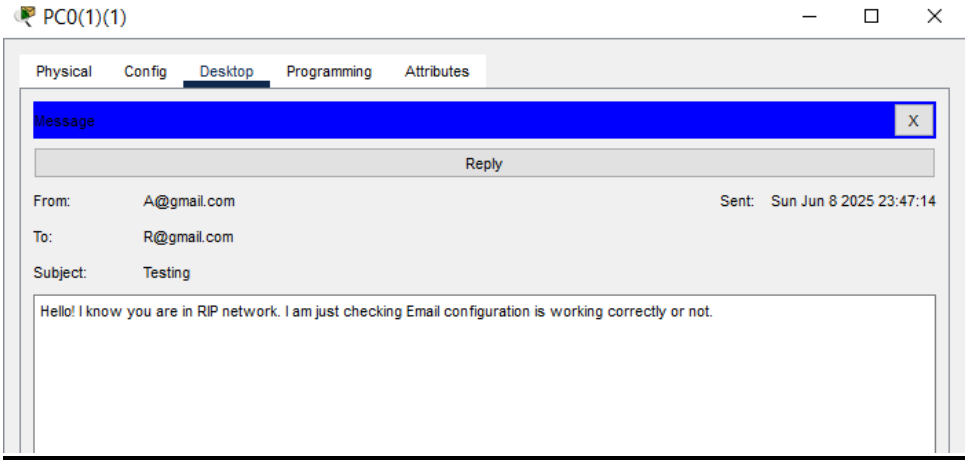


Step 4. Sending Mail to anyoune in Network.

When I compose Emial and send it to R@gmail.com(PC in RIP network) it is send successfully from PC in EIGRP.

Step 5 : Checking whether R@gmail.com(PC in RIP network) get the Email or not.

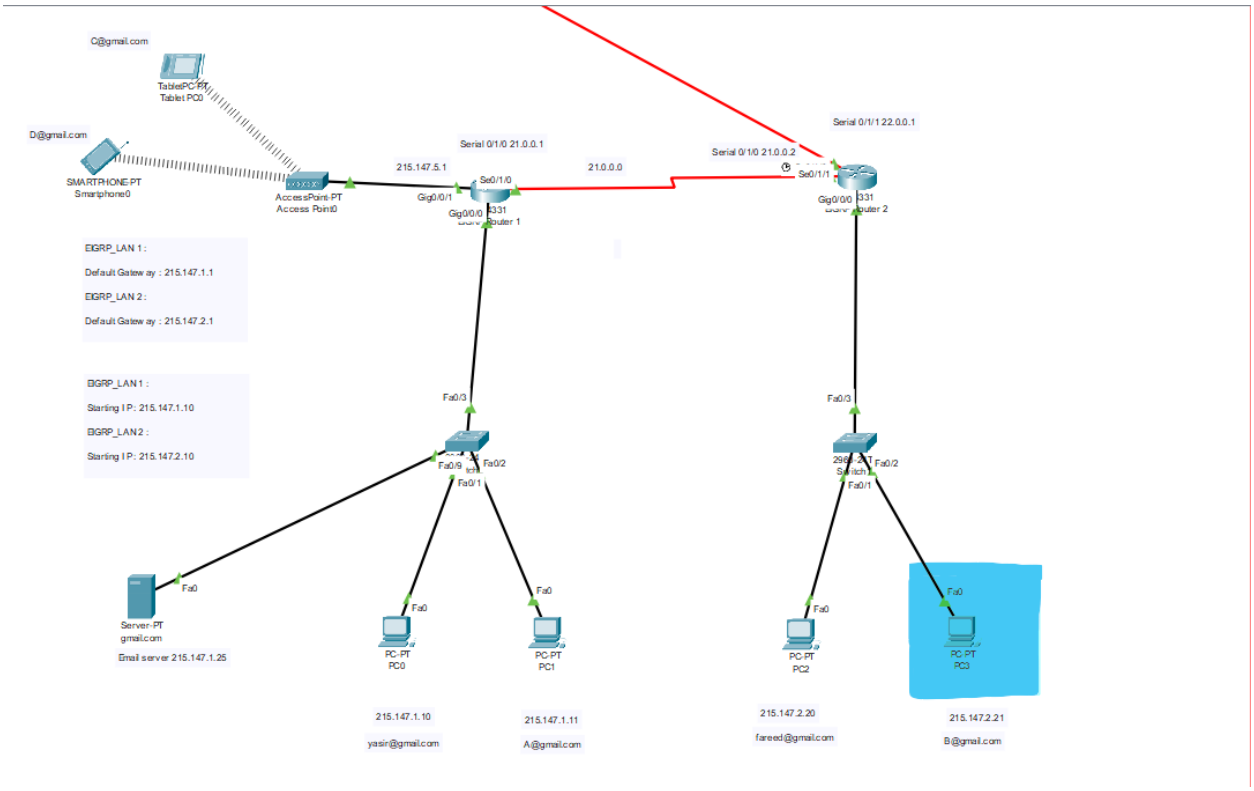




The Email is received by the PC(R@gmail.com)That means our Email configuration is correct and is working fine.

ACL (Blocking one user PC from getting message from other users): (Stage 5)

Suppose we want to avoid this highlighted PC from getting message from other PCs like 215.147.1.10 & 215.147.1.11.



Goto router above it and write the following commands.

en

conf t

access-list 10 deny host 215.147.1.10

access-list 10 deny host 215.147.1.11

(it means deny any traffic that comes to this PC from host 215.147.1.10 & 215.147.1.11)

access-list 10 permit any

(it means allow any traffic that comes from PCs other than host 215.147.1.10 & 215.147.1.11)

int serial0/1/1

ip access-group 10 in

(as the traffic is coming from 215.147.1.10 & 215.147.1.11 through serial 0/1/1 as it is inwars coming route that's why we write it.)

EIGRP Router 2

Physical Config CLI Attributes

IOS Command Line Interface

```
Press RETURN to get started!


%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0/0, changed state to up

%LINK-5-CHANGED: Interface Serial0/1/0, changed state to up

%LINK-5-CHANGED: Interface Serial0/1/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/1/1, changed state to up

%DUAL-5-NBRCHANGE: IP-EIGRP 10: Neighbor 22.0.0.2 (Serial0/1/1) is up: new adjacency

%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/1/0, changed state to up

%DUAL-5-NBRCHANGE: IP-EIGRP 10: Neighbor 21.0.0.1 (Serial0/1/0) is up: new adjacency


Router>
Router>
Router>
Router>en
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#acc
Router(config)#access-list 10 host 215.147.1.10
                                  ^
% Invalid input detected at '^' marker.

Router(config)#access-list 10 deny host 215.147.1.10
Router(config)#access-list 10 deny host 215.147.1.11
Router(config)#
Router(config)#
Router(config)#acc
Router(config)#access-list 10 permit any
Router(config)#
Router(config)#
```

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Top

EIGRP Router 2

Physical Config CLI Attributes

IOS Command Line Interface

```
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0/0, changed state to up

%LINK-5-CHANGED: Interface Serial0/1/0, changed state to up

%LINK-5-CHANGED: Interface Serial0/1/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/1/1, changed state to up

%DUAL-5-NBRCHANGE: IP-EIGRP 10: Neighbor 22.0.0.2 (Serial0/1/1) is up: new adjacency

%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/1/0, changed state to up

%DUAL-5-NBRCHANGE: IP-EIGRP 10: Neighbor 21.0.0.1 (Serial0/1/0) is up: new adjacency


Router>
Router>
Router>
Router>en
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#acc
Router(config)#access-list 10 host 215.147.1.10
                                  ^
% Invalid input detected at '^' marker.

Router(config)#access-list 10 deny host 215.147.1.10
Router(config)#access-list 10 deny host 215.147.1.11
Router(config)#
Router(config)#
Router(config)#acc
Router(config)#access-list 10 permit any
Router(config)#
Router(config)#int serial0/1/1
Router(config-if)#ip access-group 10 in
Router(config-if)#
```

Copy Paste

Top

After that when we try to reach blocked PC from PC(215.147.1.10) & from PC(215.147.1.11).

```
C:\>ping 215.147.2.21

Pinging 215.147.2.21 with 32 bytes of data:

Reply from 21.0.0.2: Destination host unreachable.
Reply from 21.0.0.2: Destination host unreachable.
Reply from 21.0.0.2: Destination host unreachable.
Reply from 21.0.0.2: Destination host unreachable.

Ping statistics for 215.147.2.21:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>
```

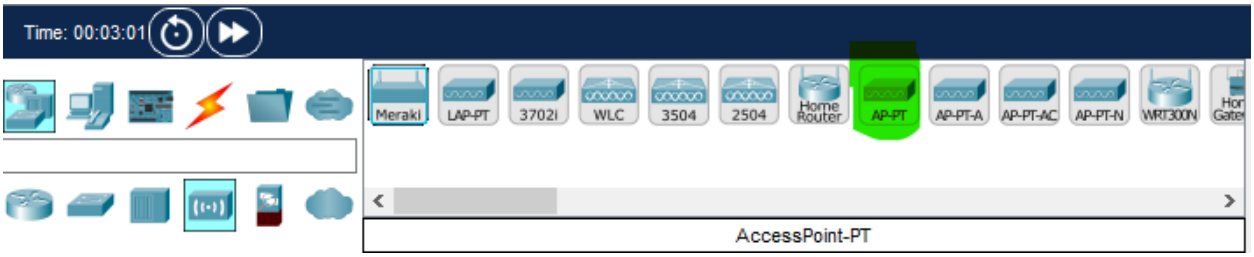
```
C:\>ping 215.147.2.21

Pinging 215.147.2.21 with 32 bytes of data:

Reply from 21.0.0.2: Destination host unreachable.
Reply from 21.0.0.2: Destination host unreachable.
Reply from 21.0.0.2: Destination host unreachable.
Reply from 21.0.0.2: Destination host unreachable.

Ping statistics for 215.147.2.21:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>
```

It says Destination Host Unreachable so, our ACL is implemented successfully.

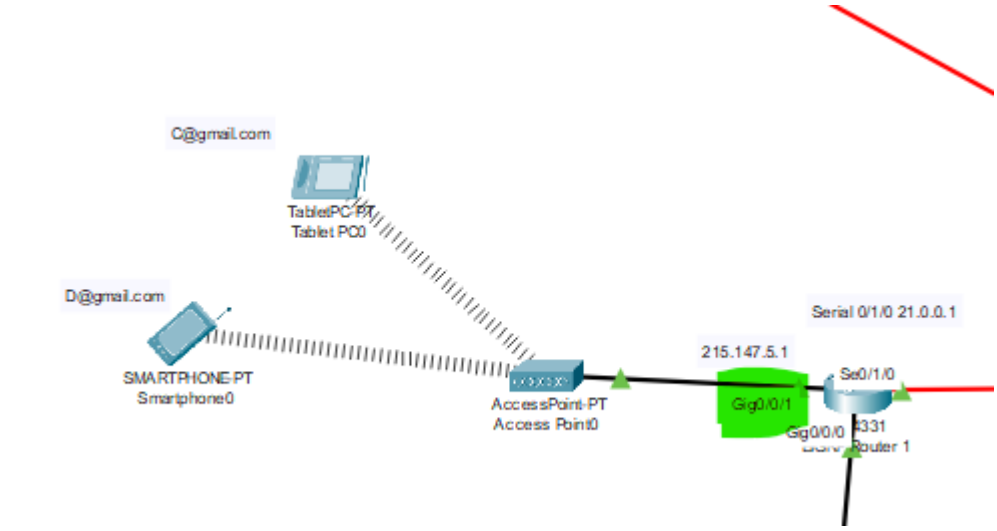
How to connect Access Point with network : (Stage 5)

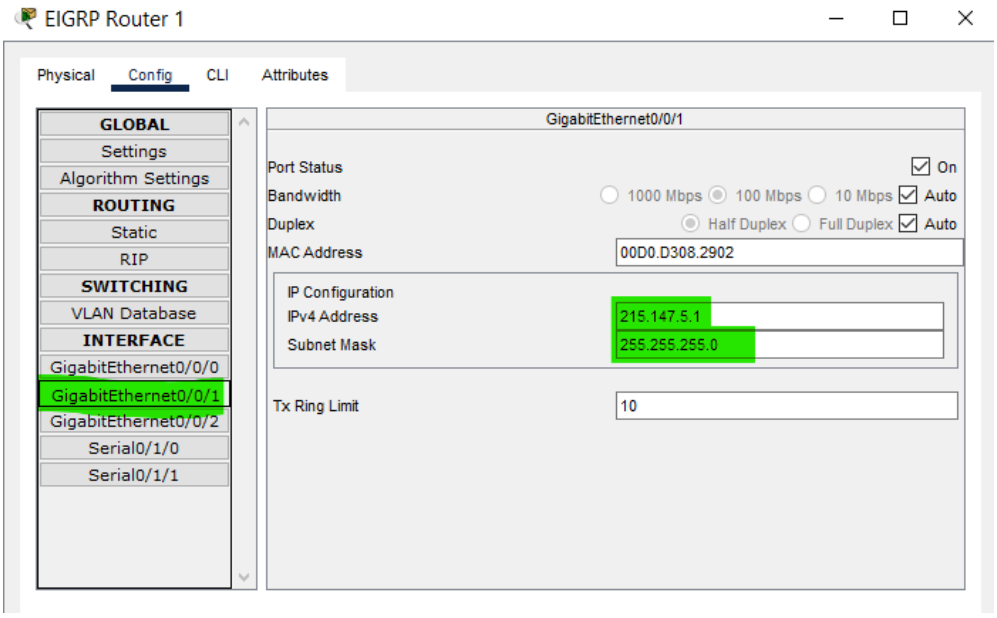
Step 1:

Place an access point from Network Devices.



Go to router and write its default gateway of network in the gig port it is connected to





Step 2:

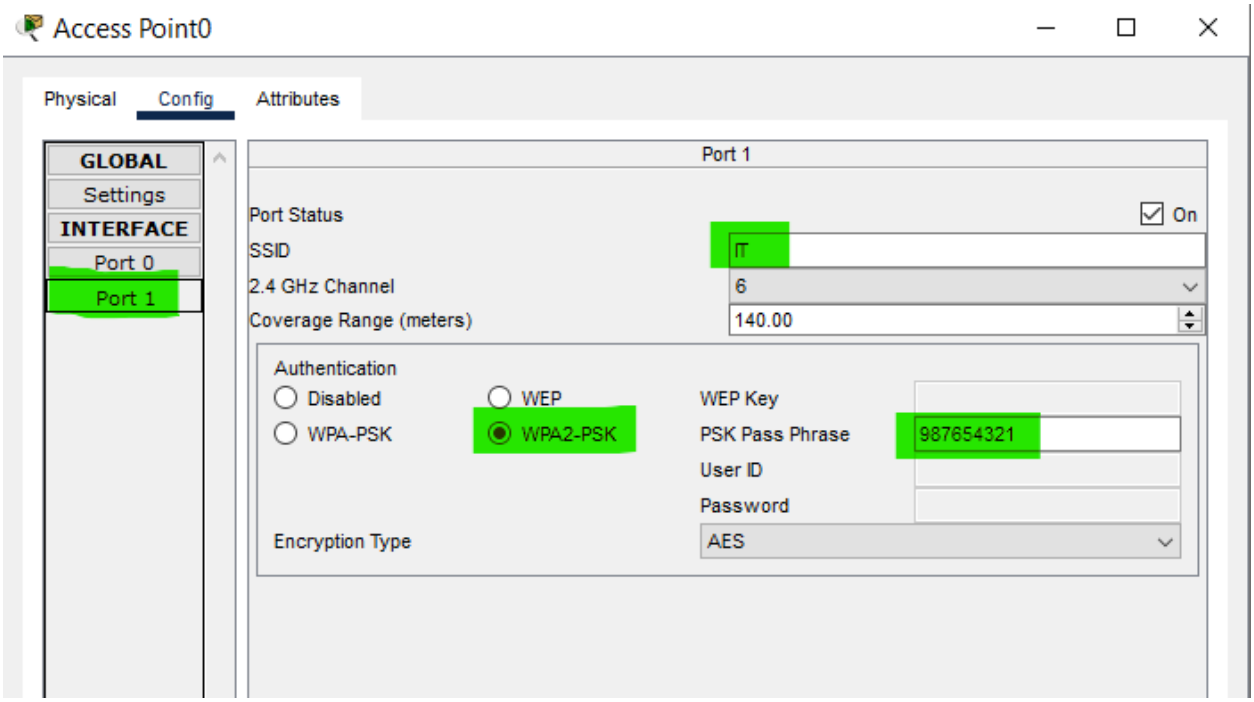
Click on access point device.

Goto Config ,

Select Port 1

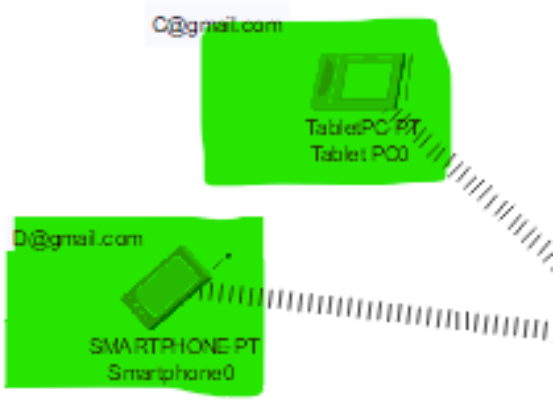
Write the SSID name **(Choose any name)**

Select WPA2-PSK **(for security)** and write the password**.**



Step 3:

Attach any Wireless Device to access point like smartphone and tablet.



Step 4:

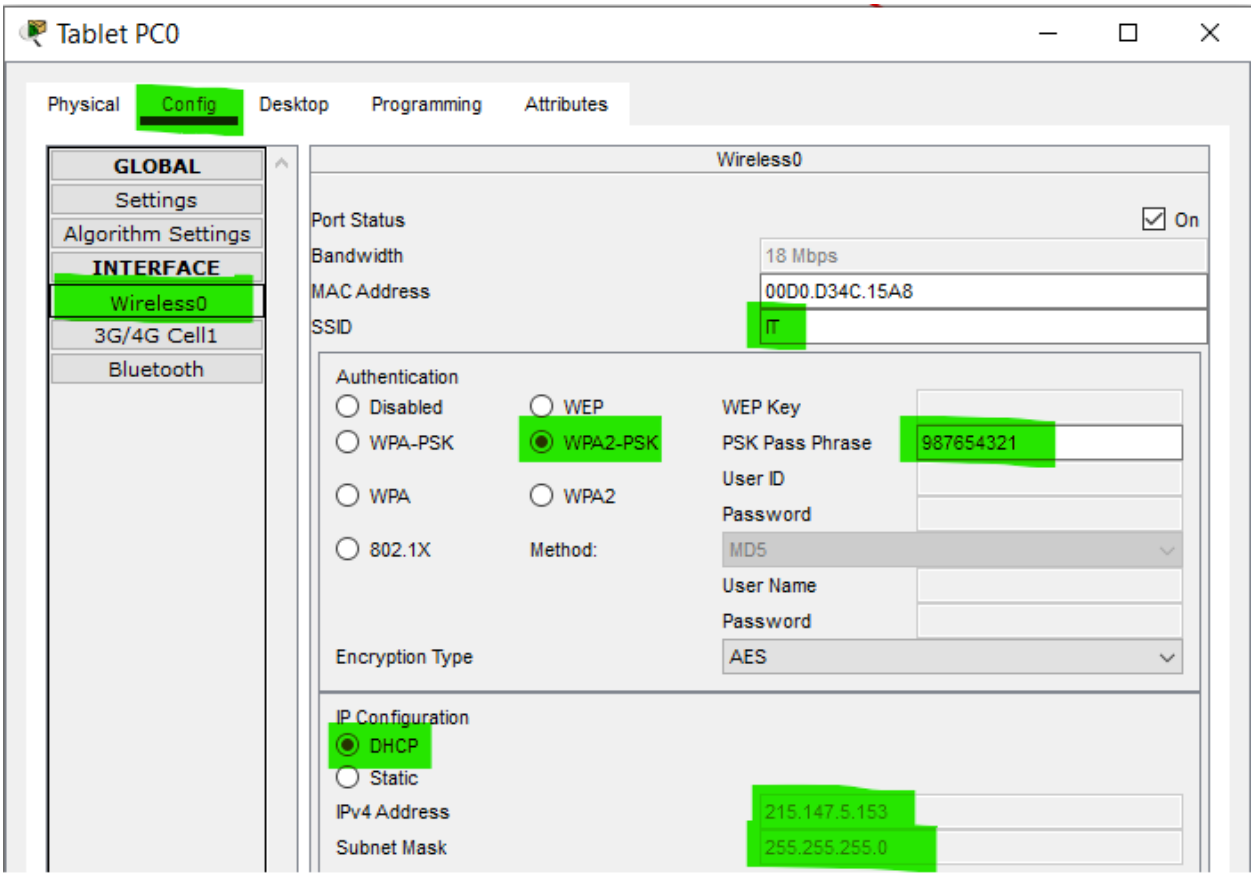
Click on wireless device and goto config

Select Wireless0

Write SSID name same **(IT)**

Select WPA2-PSK

Enter the password as well **(987654321)**



For Assigning of IP Address from DHCP server :

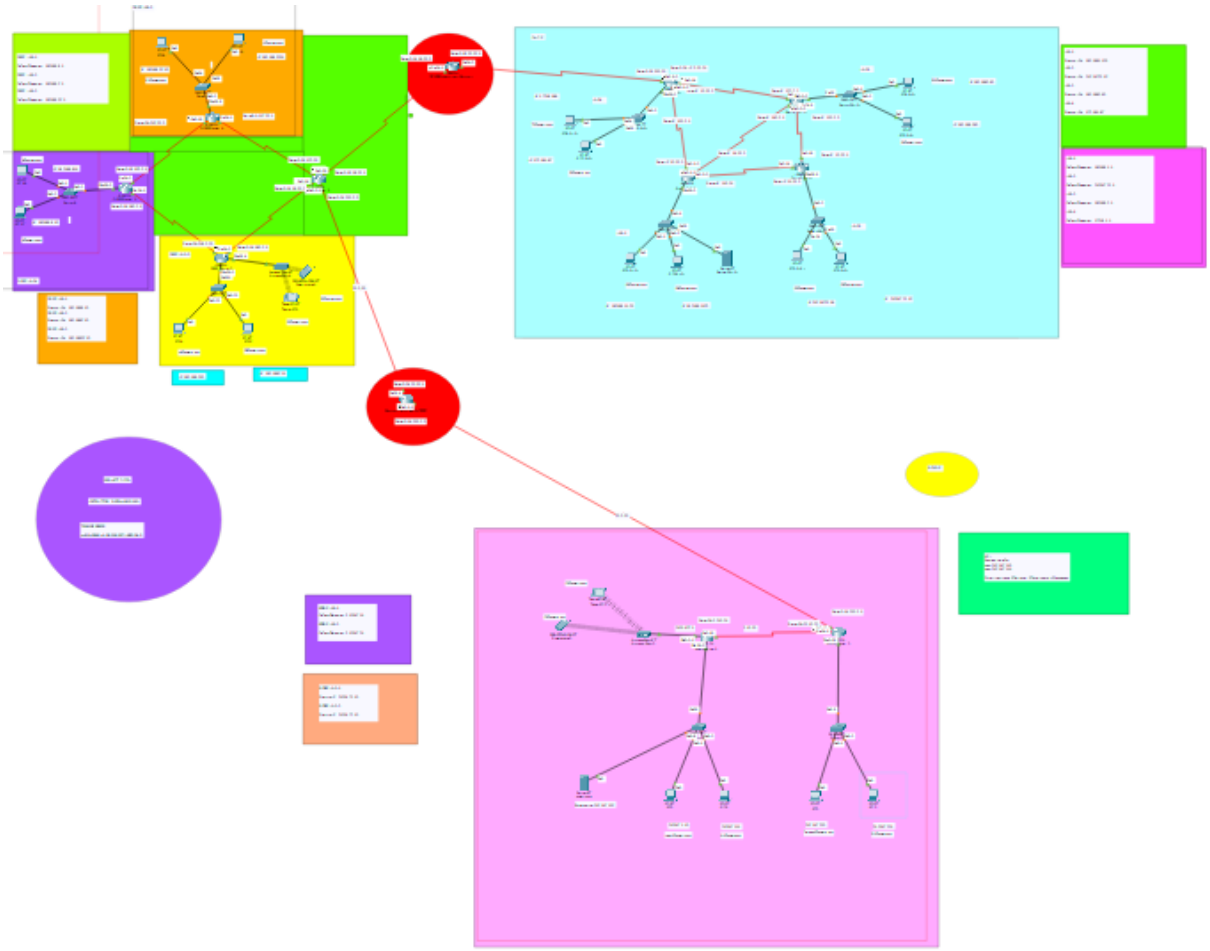
Go to Router in which access point is connected. IN CLI write the commands as

en

conf t

int gig0/0/1

ip helper-address 192.168.1.3

Also make pool for assigning IP addresses to Access Point in DHCP the click on DHCP in wireless device and it will automatically get IP address from DHCP server.

Final Topology :